

Retinal input integration in excitatory and inhibitory neurons in the mouse superior colliculus *in vivo*

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Carolyn Gehr, Jérémie Sibille, Jens Kremkow

Neuroscience Research Center, Charité-Universitätsmedizin Berlin, Charitéplatz 1, 10117 Berlin • Bernstein Center for Computational Neuroscience Berlin, Philippstraße 13, 10115 Berlin • Institute for Theoretical Biology, Humboldt-Universität zu Berlin, Philippstraße 13, 10115 Berlin • Einstein Center for Neurosciences Berlin, Charitéplatz 1, 10117 Berlin

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Abstract

The superior colliculus (SC) is a midbrain structure that contains one of the highest densities of inhibitory neurons in the brain and, together with the thalamocortical visual system, it plays a key role in visually guided behaviors. The SC receives direct inputs from retinal ganglion cells (RGCs) but whether excitatory and inhibitory SC neurons differentially integrate retinal activity *in vivo* is still largely unknown. We recently established an extracellular recording approach using high-density electrodes to measure the activity of RGCs simultaneously with their postsynaptic SC targets *in vivo*, that allows addressing how SC neurons integrate RGC activity. Here, we employ this method to study the functional properties and dynamics that govern retinocollicular signaling in a cell-type specific manner by identifying GABAergic SC neurons using optotagging in anesthetized VGAT- Chr2 mice. We measured 305 monosynaptically connected RGC-SC pairs, out of which approximately one third of retinal afferents connect onto inhibitory SC neurons. We show that both excitatory and inhibitory SC neurons receive comparably strong RGC inputs, with functionally similar RGC-SC pairs showing stronger connections. Our results demonstrate that similar wiring rules apply for RGCs innervation of both excitatory and inhibitory SC neurons, which is unlike the cell-type specific connectivity in the thalamocortical system. Contrasting the similar RGC-SC connection strength, we observed that RGC activity contributed more to the activity of postsynaptic excitatory SC neurons than to the activity of postsynaptic inhibitory SC neurons. This implies that the excitatory SC neurons are more specifically coupled to RGC afferent inputs, while inhibitory SC neurons may integrate additional inputs from other sources. Taken together, our study deepens the understanding of cell-type specific retinocollicular functional connectivity and emphasizes that the two major brain areas for visual processing, the visual cortex and the superior colliculus, differently integrate sensory afferent inputs.

eLife assessment

This **valuable** study contributes to understanding how retinal activity shapes the response properties of excitatory and inhibitory neurons in a major visual target, the superior colliculus. The evidence supporting the claim is **convincing**: the work is technically excellent yet the interpretation of these results assumes an unbiased sampling and integration of the RGC axon in the SC, a caveat pointed out by the authors. Overall, this study provides insights into the integration of visual information from the eye to the brain, and this work will be of interest to visual neuroscientists.

Introduction

The mouse superior colliculus (SC) is a midbrain structure that receives direct inputs from retinal ganglion cells (RGCs) (Dräger and Hubel, 1976 [↗](#); Ellis et al., 2016 [↗](#); Huberman et al., 2008 [↗](#); Kremkow and Alonso, 2018 [↗](#)). Together with the visual cortex (Glickfeld et al., 2013 [↗](#); Lashley, 1931 [↗](#); Niell and Scanziani, 2021 [↗](#); Petruno et al., 2013 [↗](#)), the SC plays a key role in visually guided behaviors (Evans et al., 2018 [↗](#); Hoy et al., 2019 [↗](#); Shang et al., 2019 [↗](#), 2015 [↗](#); Wei et al., 2015 [↗](#)). Intriguingly, the visual SC layers contain one of the highest densities of inhibitory (GABAergic) neurons in the brain (Kaneda et al., 2008 [↗](#); Ranney Mize, 1992 [↗](#)) suggesting that inhibition plays a key role in visual processing within the SC. Indeed, inhibitory SC neurons (INs) are known to be involved in several sensory functions including surround suppression (Kasai and Isa, 2016 [↗](#)) and motion processing (Barchini et al., 2018 [↗](#); Gale and Murphy, 2016 [↗](#)), but also in the regulation of wakefulness (Zhang et al., 2019 [↗](#)). However, how GABAergic SC neurons are recruited by their retinal inputs *in vivo* remains largely unknown (Shi et al., 2017 [↗](#)). An understanding of how RGC inputs are integrated by SC INs and SC excitatory neurons (EXNs) is crucial for reaching a mechanistic understanding of the computations within the SC microcircuit.

In sensory systems, the divergence of long-range afferent axons onto GABAergic and non-GABAergic neurons ensures a balance between excitation and inhibition (E/I) (Miller, 2016 [↗](#)). For example, in sensory cortices, thalamic afferents differentially activate excitatory and local inhibitory neurons (Cruikshank et al., 2007 [↗](#); Ji et al., 2016 [↗](#)) with a stronger drive onto inhibitory neurons compared to excitatory neurons (Bruno and Simons, 2002 [↗](#); Cruikshank et al., 2007 [↗](#); Gabernet et al., 2005 [↗](#); Gibson et al., 1999 [↗](#); Inoue and Imoto, 2006 [↗](#); Swadlow, 2003 [↗](#)). This stronger drive on inhibitory neurons establishes an effective feedforward inhibition (Agmon and Connors, 1992 [↗](#); Bereshpolova et al., 2020 [↗](#); Gabernet et al., 2005 [↗](#); Kremkow et al., 2010a [↗](#), 2010b [↗](#); Sun et al., 2006 [↗](#)) that balances the excitation from thalamic afferents (Isaacson and Scanziani, 2011 [↗](#)) and contributes to sharpening stimulus feature selectivity (Lee et al., 2012 [↗](#)). Moreover, this wiring motif can partly be linked to the response properties of cortical inhibitory neurons as they typically show higher firing rates, broader tuning, low selectivity and high sensitivity (Alonso and Swadlow, 2005 [↗](#); Bruno and Simons, 2002 [↗](#); Gibson et al., 1999 [↗](#); Kremkow et al., 2016 [↗](#); Porter et al., 2001 [↗](#); Swadlow et al., 2002 [↗](#)) to facilitate feedforward inhibition in response to thalamic input. Despite the importance of this afferent wiring motif for sensory processing in the thalamocortical visual circuit, it is currently unknown whether the retinocollicular pathway follows similar or different principles.

We recently demonstrated that SC neurons can receive strong input from RGC axons *in vivo* (Sibille et al., 2022a [↗](#)). Moreover, around one third of retinocollicular synapses connect onto inhibitory neurons that form intrinsic connections within the SC (Whyland et al., 2020 [↗](#)). These findings

support the notion that an effective feedforward inhibition might also be at play in the retinocollicular system.

But whether afferent inputs are differentially integrated by excitatory and inhibitory SC neurons, similar to the thalamocortical system, is unknown. However, our previous study in the retinocollicular system lacks the cell-type specificity, and it remains an open question whether the strong drive is specific to inhibitory SC neurons or if it is a general property of retinocollicular connections, and thus cell-type independent. Therefore, whether there are any differences in connection strengths between RGCs that project to excitatory and inhibitory SC neurons remains elusive. In this study, we investigate how excitatory and inhibitory neurons in the SC integrate retinal inputs *in vivo*.

We aim to answer these questions by characterizing the properties of the long-range synaptic connections between RGC-SC pairs employing a set of measurements: 1. The connection efficacy is used to study how strongly excitatory and inhibitory SC neurons are driven by their RGC input. 2. The functional similarity is analyzed between connected RGC-SC pairs to examine whether comparable wiring motifs govern retinocollicular signaling in excitatory and inhibitory neurons. 3. Short-term dynamics of retinal activity are studied to estimate whether the retinal spiking pattern shows paired-spike enhancement on retinocollicular synapses. Finally, 4. the connection contribution is employed to investigate how tightly the SC activity is coupled to retinal spiking.

This study intends to generate a more detailed picture of cell-type specific retinocollicular innervation and to fill a gap concerning the different long-range functional connectivity motifs that exist along the visual pathways.

Results

Recording excitatory and inhibitory neurons simultaneously with RGC axons in the mouse SC *in vivo*

To investigate the integration of retinal input in excitatory and inhibitory neurons in the SC *in vivo*, we recorded extracellular neural activity in the visual layers of SC in VGAT-ChR2 mice using high-density Neuropixels probes (Jun et al., 2017 [↗](#)). We employed a tangential insertion approach to maximize the extent of visually driven channels in the SC (**Figure 1A** [↗](#)) (Sibille et al., 2022a [↗](#)). Notably, we recently demonstrated that this approach allows us to not only measure spiking activity from SC neurons, but also to record electrical signals from retinal ganglion cell axons (RGC) that terminate in the SC (Sibille et al., 2022a [↗](#)). The somatic action potentials of SC neurons can be distinguished from the axonal action potentials of RGCs based on their spatiotemporal waveforms profile (Sibille et al., 2022a [↗](#)). RGC axonal waveforms are triphasic with a larger spatial spread (**Figure 1D** [↗](#) left) while somatic SC neurons show a characteristic biphasic waveform profile (**Figure 1D** [↗](#), middle and right).

To study cell-type specific differences, we combined the Neuropixels recordings with an optogenetic identification strategy (optotagging) to identify GABAergic SC neurons in VGAT-ChR2-EYFP (Zhao et al., 2011 [↗](#)) mice (**Figure 1A** [↗](#), B; data from $n = 11$ experiments in $n = 9$ mice included in the analysis, if not stated otherwise). In these transgenic mice, GABAergic neurons specifically express Channelrhodopsin-2 and can be identified upon blue light stimulation. To photo-activate a large proportion of GABAergic SC neurons using short light pulses (100ms pulses, 2Hz, 2.7 mW) we inserted the optic fiber perpendicularly to the Neuropixels probe (**Figure 1A** [↗](#), right, see Material and Methods). Using this recording and light stimulation configuration we could activate GABAergic SC neurons across an average of ~ 1.5 mm of SC tissue (light-activated range = 1.50 ± 0.41 mm, **Figure 1B** [↗](#)) which also allowed us to record SC neurons along continuous retinotopy (**Figure 1C** [↗](#), right) in a cell-type specific manner (**Figure 1G** [↗](#)). We used

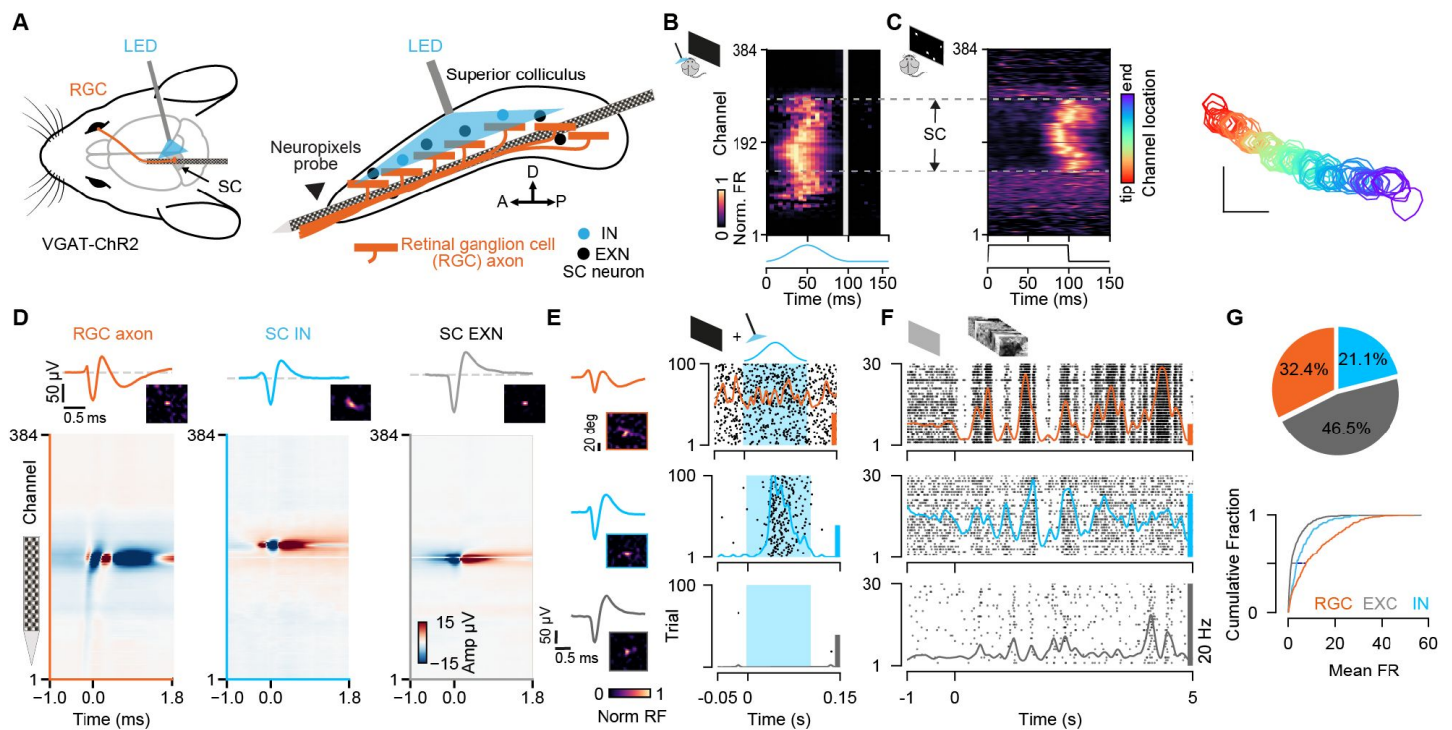


Figure 1

Simultaneous extracellular recordings of RGC axons and SC neurons combined with optotagging identifies GABAergic neurons in VGAT-ChR2 mice.

A, Recording configuration for tangential electrode insertion and optotagging in the visual layers of mouse SC. The optogenetic fiber is inserted perpendicularly to the Neuropixels probe to activate GABAergic cells in VGAT-ChR2 mice. RGC afferents (orange) project onto GABAergic (blue) and non-GABAergic (black) neurons in SC. **B**, Multi-unit response (MUA) to optogenetic stimulation along the 384 recording sites during the presentation of a black screen. The large spatial extent of the optogenetic activation is evident. Gray vertical bar = light artifacts induced by the LED stimulation. **C**, Visually evoked MUA during the presentation of a sparse noise stimulus along the recording sites. Receptive field contours of recording sites with high signal-to-noise ratio. The color code reflects the location within the SC. Scale bar represents 10 degree. **D**, Spatiotemporal waveform profiles. Single channel waveforms identified at the peak channel (top) and multi-channel waveforms (bottom) for RGC axon (left, orange) and inhibitory (middle, blue) and excitatory (right, gray) SC neurons. Receptive fields (RFs) indicate visually responsive neurons. **E**, Identification of SC cell-types via optotagging. Raster plots and peri-stimulus time histograms (PSTHs) for single-neuron responses to blue light pulses (100ms) presented under baseline conditions. Excitatory SC neurons (EXNs, bottom) and RGCs (top) do not respond to the LED pulse, while GABAergic SC neurons respond to the light pulse with an increase in spiking response (middle). Optogenetic stimulation period is highlighted in blue. The colored scale bars on the right represent 20 Hz firing rate. **F**, PSTHs and raster plots for different cell types shown in E in response to a natural movie stimulus (10s, 30 trials). Note the high firing rate in RGCs. **G**, Top: Proportion of identified GABAergic (INs, blue), non-GABAergic (EXNs, gray) SC neurons and retinal axons (RGCs, orange) populations. Note that around one third of the captured SC neurons are GABAergic (n = 326 RGCs, n = 468 EXNs, n = 212 INs). Bottom: Cumulative distribution of mean firing rates in response to a natural movie stimulus presented for 10s, 30 trials. Two-sided Wilcoxon rank-sum test.

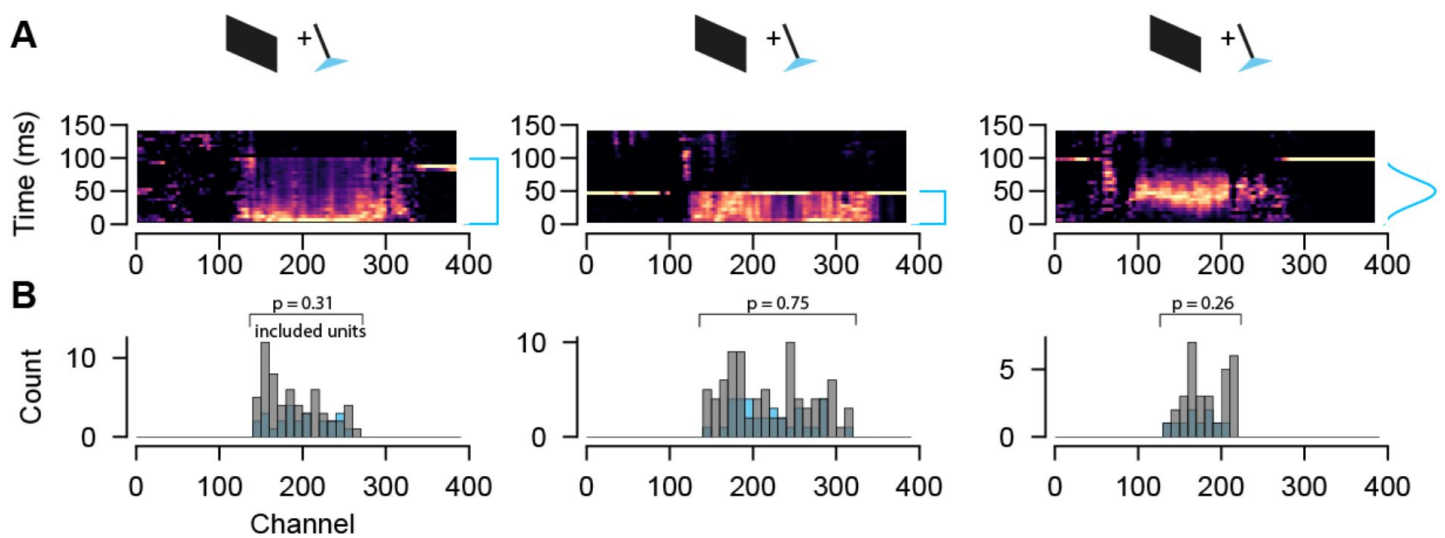


Figure 1

figure supplement 1 Optogenetic response and spatial distribution of excitatory and inhibitory SC neurons.

A, Multi-unit activity in response to optogenetic stimulation (100 or 50 ms) using a square wave pulse or a Gaussian-shaped LED pulse for three individual recordings. **B**, Spatial distribution of included excitatory and inhibitory SC single units within the range of channels that are activated via LED stimulation. Both cell types are equally distributed. Two-sided Wilcoxon rank-sum test.

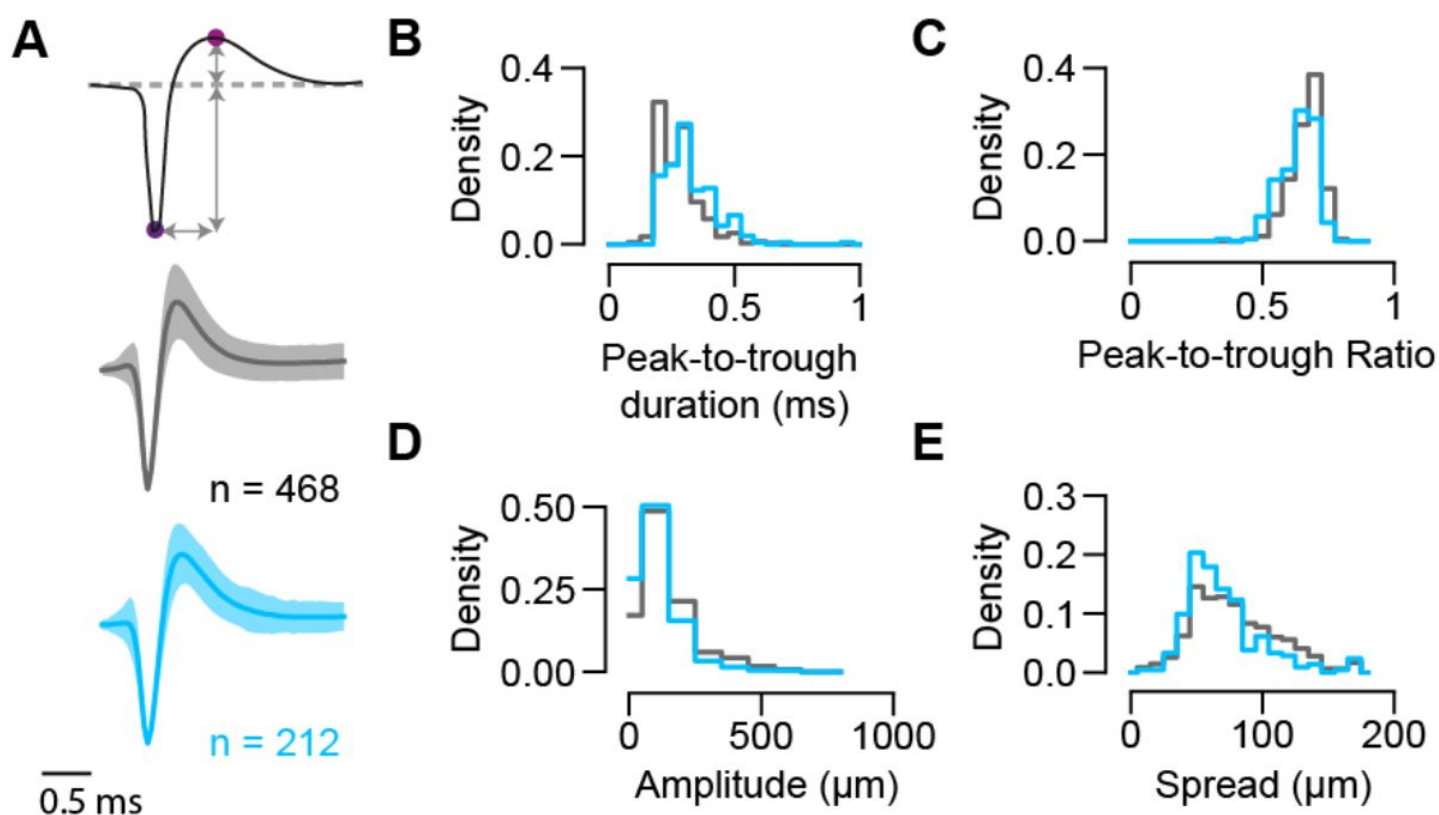


Figure 1

figure supplement 2 Spike waveform features analysis for GABAergic and non-GABAergic neuron populations in the SC.

A, Top: Illustration of features extracted from single-channel waveforms (magenta circles indicate the trough and peak). Middle and bottom: Single-channel mean waveforms for non-GABAergic (gray; n = 468) and GABAergic (blue, n = 212, n = 9 mice) SC populations (mean±SEM). **B-D**, Distribution of waveform features extracted from single-channel waveforms for inhibitory and excitatory SC populations. Peak-to-trough duration represents the time between trough and peak ($p = 4.05 \times 10^{-10}$). Peak-to-trough ratio represents the ratio between amplitudes of peak and trough ($p = 4.45 \times 10^{-9}$). Amplitude is the absolute difference between trough and peak ($p = 1.38 \times 10^{-9}$). **E**, Distribution of waveform spread along the probe extracted from multichannel waveforms for GABAergic and non-GABAergic SC populations ($p = 1.58 \times 10^{-4}$, n = 468 EXNs, n = 212 INs, Two-sided Wilcoxon rank-sum test).

the visually evoked activity and continuous retinotopy to demarcate the boundaries of the SC (Figure 1C). Moreover, for the characterization of excitatory and inhibitory classes of SC neurons, we limited our analysis to single units that were localized within the range of optogenetic stimulation (Figure 1B/C and Figure 1 – figure supplement 1A, see Material and Methods). Cell types were characterized using a custom-written graphic-user-interface (GUI, see Material and Methods) to label units that were activated by the LED pulse as inhibitory neurons (INs, Figure 1E, middle). These neurons were distinguished based on their reliable increase in firing rate in response to optogenetic stimulation during spontaneous (black screen) and visually evoked conditions (checkerboard stimulus, see Material and Methods). SC neurons that were not modulated by the LED pulses, but located within the light-activated range, were labeled as excitatory (EXNs, Figure 1E bottom). Excitatory and inhibitory SC neurons were equally distributed within the boundaries of LED stimulation for 9/11 experiments (p values = 3.07×10^{-1} , 1.15×10^{-2} , 7.55×10^{-1} , 8.34×10^{-1} , 5.01×10^{-6} , 0.79, 0.80, 0.26, 0.33, 0.08, 0.13, Wilcoxon rank-sum test). In only two experiments EXNs and INs were not equally distributed, but only 2 single units were located in the region without included SC IN units, that passed the quality criteria for spike sorting (Figure 1 – figure supplement 1B, see Material and Methods). As expected, retinal axons were not activated by the LED pulse (Figure 1E, top), except for a few cases (4/326 RGCs) which were excluded from the connectivity analysis in this study. In total, we recorded 326 RGC axons and 680 SC neurons. Among the recorded SC neurons one third (31.2%) were GABAergic (Figure 1G, $n = 468$ EXNs, $n = 212$ INs).

Having sorted the SC neurons into the cell classes using optogenetic stimulation (EXNs and optogenetically identified INs) we compared different spike waveform features to test whether SC neurons show differences in their waveforms. Characterizing the spike waveforms is commonly used in the cortex or hippocampus to differentiate between glutamatergic and GABAergic neurons (Lee et al., 2016; Moore and Wehr, 2013; Niell and Stryker, 2008). However, the separation based on differences in waveform features was not an adequate measure to separate SC EXNs from SC INs (Figure 1 – figure supplement 2). This result is in line with findings by Essig et al. (2021) who did not observe a cell type specific difference between waveform features in the SC (Essig et al., 2021). Similar findings have been shown in the inferior colliculus (Ono et al., 2017), a neighboring midbrain structure, suggesting that waveform classification analyses used in the cortex are not suitable for distinguishing neurons in the midbrain.

To further characterize the response properties of the different cell classes (RGC, SC INs, SC EXNs), we measured their firing rates (FR) in response to a natural movie stimulus provided in Froudarakis et al. (2014) (Froudarakis et al., 2014) (see Material and Methods). We found that RGCs show the highest FR (firing rate RGC: median = 7.55 spikes/s, $Q1 = 2.39$ spikes/s, $Q3 = 17.93$ spikes/s, $n = 326$ EXNs; Figure 1G, bottom). On average, the mean FR in SC INs (firing rate IN: median = 3.63 spikes/s, $Q1 = 1.6$ spikes/s, $Q3 = 8.02$ spikes/s, $n = 212$ INs) was 1.8 times higher compared to that of SC EXNs (firing rate EXN: median = 1.38 spikes/s, $Q1 = 0.49$ spikes/s, $Q3 = 4.19$ spikes/s, $n = 468$ INs; SC EXN vs SC IN $p = 3.1567 \times 10^{-15}$, SC EXN vs RGC $p = 4.8362 \times 10^{-40}$, SC IN vs RGC $p = 2.7388 \times 10^{-8}$, two-sided Wilcoxon rank-sum test). Our results are in line with previous studies by Reid (1998, 2003) showing that RGCs show high mean firing rates (Kara and Reid, 2003; Usrey et al., 1998). Moreover, from other midbrain areas it is also known that GABAergic neurons show higher FRs compared to glutamatergic cells (Ono et al., 2017).

In summary, by combining Neuropixels recordings to measure the activity of RGC axons and SC neurons with optogenetic identification of GABAergic SC neurons, it becomes possible to measure monosynaptic retinocollicular connectivity *in vivo* and retinal innervation strength in a cell-type specific manner.

Retinal input integration in excitatory and inhibitory SC neurons

To study the monosynaptic retinocollicular connectivity *in vivo* and retinal innervation strength, we identified putative monosynaptically connected RGC-SC pairs employing established cross-correlation analysis (CCG) methods (Alonso et al., 2001 [↗](#); Bereshpolova et al., 2020 [↗](#); Sibille et al., 2022a [↗](#)). A significant transient and a short-latency peak in the spike train CCG is a hallmark of monosynaptic connectivity (**Figure 2A** [↗](#)) and identifies connected RGC-SC pairs. We measured up to 98 monosynaptic connections in individual recordings, depending on the number of measured RGC axons (**Figure 2C** [↗](#)). In total, we identified 305 connected RGC-SC pairs, of which 29.8% were retinal axons that connected onto GABAergic SC neurons ($n = 214$ connected RGC-SC EXN pairs, $n = 91$ connected RGC-SC IN pairs, **Figure 2B** [↗](#)/D). This high yield in monosynaptically connected pairs was due to the close proximity of the simultaneously recorded RGC axons and SC neurons on the high-density electrode (**Figure 2D** [↗](#), bottom) (Sibille et al., 2022a [↗](#)). For most of the connected RGC-SC pairs the peak channels on the electrode were on average 50–60 μm apart on the probe and hence within SC (distance RGC-EXN: median = 40.0 μm , first quartile = 25.61 μm , third quartile = 62.1 μm ; distance RGC-IN: median = 51.22 μm , first quartile = 32.0 μm , third quartile = 83.08 μm). We found a significant difference in peak channel distance between RGC-SC EXN and RGC-SC IN pairs with longer distances between RGC axons and inhibitory SC neurons ($p = 0.03287$, two-sided Wilcoxon rank-sum test), however, the effect size was small (Cohen's $d = 0.0915$). The maximum distance for retinal axons and inhibitory SC neurons was 240 μm , while the peak channels of one RGC-SC EXN pair were 960 μm apart. The latter pair might include a SC wide field neuron as these cells are excitatory and their dendritic arbor widths have been reported to spread up to 900 μm (Gale and Murphy, 2014 [↗](#)).

To assess the connection strength between RGCs and excitatory and inhibitory SC neurons, we calculated the connection efficacy (Usrey et al., 1999 [↗](#)). The efficacy measures the connection strength as the probability of an RGC spike triggering an action potential in the postsynaptic SC neuron (see Material and Methods). We found highly diverse connectivity patterns for RGC-SC EXC and RGC-SC IN pairs. RGC activity evoked robust firing in both SC cell types (**Figure 2E** [↗](#), top), however, pairs with weak connection strength were also observed as recently described in Sibille et al. (2022a [↗](#)) (**Figure 2E** [↗](#), bottom). Across the population, both SC cell types received similarly strong input from the retina (efficacy RGC-SC IN: median = 4.54 %, $Q1 = 2.2$ %, $Q3 = 7.96$ %, maximum = 62.05 %, $n = 91$ connected pairs; efficacy RGC-SC EXN: median = 3.5 %, $Q1 = 1.96$ %, $Q3 = 5.89$ %, maximum = 41.12 %, $n = 214$ connected pairs; **Figure 2F** [↗](#)). While we observed a tendency for slightly stronger RGC-SC IN connections, this difference was not statistically significant ($p = 0.053$, two-sided Wilcoxon rank-sum test). To assess whether the connectivity strength is related to the firing of the postsynaptic cell, we examined the relationship between the efficacy and the overall firing rate throughout the entire recording session. Surprisingly, our results demonstrate a positive correlation between the efficacy and the mean firing rate on the population level (**Figure 2G** [↗](#), RGC-SC EXN $r = 0.42685$; $p < 0.0005$; RGC-SC IN: $r = 0.43543$, $p = 0.00002$, two-sided Wilcoxon rank-sum test).

Here, we set out to explore how RGC spikes drive the activity in postsynaptic excitatory and inhibitory SC neurons. We did not find a cell-type specific difference in connection strength and therefore conclude that inhibitory and excitatory SC neurons are innervated comparably strong and are both reliably driven by their retinal inputs.

Functional similarity of retinocollicular connections

In sensory cortices, inhibitory neurons are less selective and receive a diverse and nonspecific set of thalamic inputs compared to excitatory neurons which receive more specific thalamic inputs (Alonso and Swadlow, 2005 [↗](#); Bruno and Simons, 2002 [↗](#)). However, the selectivity and functional similarity between RGCs and their excitatory and inhibitory target populations in SC remains largely unknown (Shi et al., 2017 [↗](#)). Despite both SC populations showing a similar innervation strength from retinal axons, inhibitory neurons could sample from a more functionally diverse

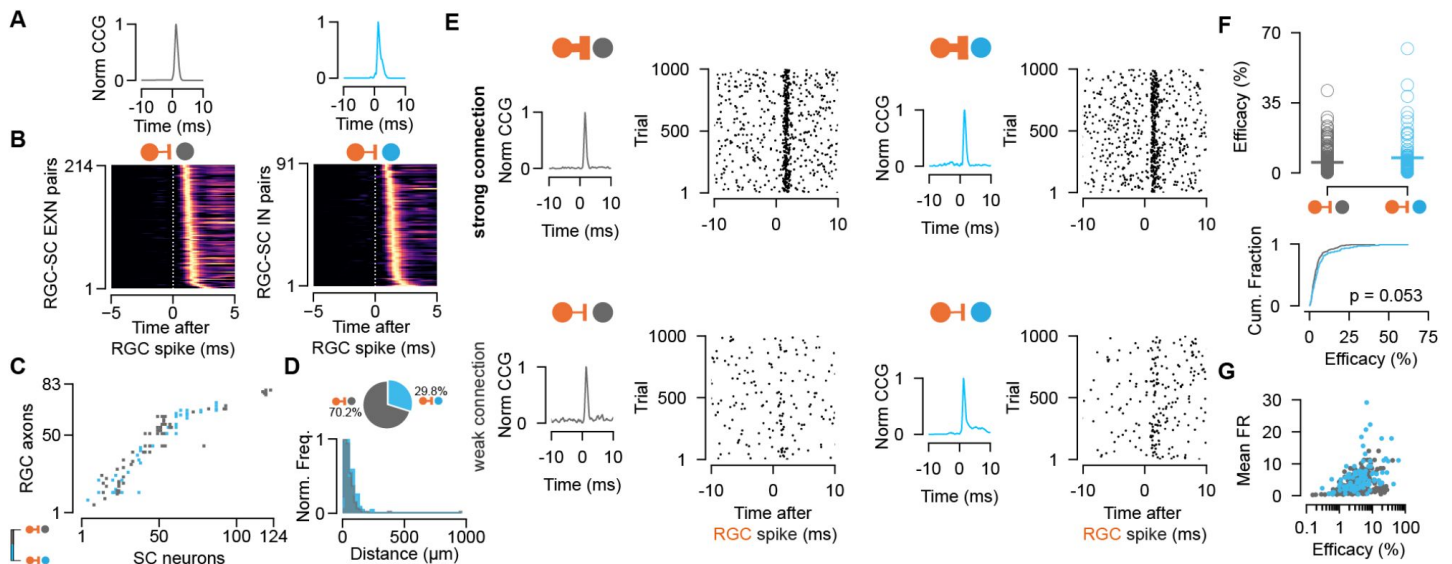


Figure 2

Retinal innervation is similarly strong to excitatory and inhibitory SC neurons.

A, Monosynaptically connected RGC-SC EXN (gray) and RGC-SC IN (blue) pairs are identified via cross-correlation analysis (CCG). **B**, CCGs of connected RGC-SC EXN and RGC-SC IN pairs sorted by their peak latency ($n = 214$ RGC-SC EXN, $n = 91$ RGC-SC IN, $n = 11$ recordings). **C**, Connectivity matrix from a single recording. Gray marks indicate connections onto excitatory SC neurons, blue marks indicate connections onto inhibitory SC neurons. RGC axons and SC neurons are sorted by their peak channel along the electrode. **D**, Distribution of peak channel distances between RGC axons and connected SC neurons ($p = 0.0328$, two-sided Wilcoxon rank-sum test). Inset shows pie chart of identified RGC-SC IN and RGC-SC EXN pairs. **E**, Elicited SC spiking in response to firing of a presynaptically connected retinal ganglion cell (RGC). Raster plot shows SC firing to 1000 randomly selected RGC spikes. Both SC cell types show robust activation upon RGC spiking (top) but also weaker connections can be found (bottom). **F**, Scatter plot and cumulative distribution of synaptic efficacy as a measure for connection strength for RGC-SC EXN and RGC-SC IN connected pairs ($p = 0.053$, $n = 214$ RGC-SC EXN, $n = 91$ RGC-SC IN). **G**, Efficacy as a function of mean FR during the entire recording session (RGC-SC EXN $r = 0.42685$; $p < 0.0005$; RGC-SC IN: $r = 0.43543$, $p = 0.00002$). Two-sided Wilcoxon rank-sum test.

population of RGCs. To test this, we characterized the functional similarity between the RGCs and their connected postsynaptic SC neurons. To this end, we calculated the correlation coefficients between the trial-averaged visually evoked activity of connected RGC-SC pairs in response to light and dark sparse noise stimuli (r_{SL} & r_{SD}) as well as the natural movie stimulus (r_{NM}), a phase scrambled version (r_{PSM}) and a checkerboard stimulus (r_{Chk}) (**Figure 3A** [↗](#)/B, RGC-SC with high signal-to-noise in their receptive field, see Material and Methods). To quantify the overall functional similarity by a single similarity value, we averaged the five correlation measurements ($\text{similarity} = (r_{SL} + r_{SD} + r_{NM} + r_{PSM} + r_{Chk})/5$). A similarity value of 1 corresponds to perfectly correlated visually driven responses between the RGC and the postsynaptic SC neuron, while a value of 0 signifies uncorrelated responses. In our data, the RGC-SC pairs span a wide range of functional similarity, however, we found no significant difference between RGC-SC EXN and RGC-SC IN pairs (similarity for RGC-SC EXN = 0.44 ± 0.15 and RGC-SC IN = 0.37 ± 0.19 , $n = 85$ RGC-SC EXN pairs, $n = 29$ RGC-SC IN pairs, $p = 0.245$, two-sided Wilcoxon rank-sum test, **Figure 3B** [↗](#)). In a next step, we correlated the connection efficacy measure to the similarity index. The synaptic efficacy was positively correlated with the functional similarity of the connected pairs, with a stronger correlation for RGC-SC IN pairs (RGC-SC EXN $r = 0.37$, $p = 0.0005$; RGC-SC IN $r = 0.44$, $p = 0.02$, **Figure 3C** [↗](#)). Overall, we observed that functionally similar pairs are more strongly connected. However, we also observed cases of relatively strong connected RGC-SC pairs (~10%) with low similarity (**Figure 3C** [↗](#)), suggesting that some SC neurons receive convergent input from a functionally more diverse pool of RGC afferents (Sibille et al., 2022a [↗](#)). Our findings delineate that both RGC-SC EXN and RGC-SC IN connected pairs are organized, while functionally similar properties between the pre-and postsynaptic neuron are reflected in a stronger connection.

Paired-spike interactions of the retinocollicular connections

We showed that retinal spikes are efficient in driving SC activity for both excitatory and inhibitory neurons. As the timing of presynaptic activity has been assessed as a parameter that modulates postsynaptic firing (Zucker, 1989 [↗](#)) and connection strength (Kara and Reid, 2003 [↗](#); Usrey et al., 1998 [↗](#)), we wanted to know whether the temporal pattern of retinal activity has different effects on the synaptic efficacy of excitatory and inhibitory SC neurons. To that end, we studied the short-term dynamics (facilitation or depression) of the measured connections and performed a paired-spike analysis approach as described by Usrey and colleagues in 1998 (Usrey et al., 1998 [↗](#)). A pair of retinal spikes (1st, 2nd RGC) was defined by its temporal firing patterns: inter-spike interval (ISI) and dead time. We selected RGC spikes for which a 1st RGC followed the 2nd RGC by a specific ISI (minimum ISI = 5ms, maximum ISI = 30ms). The dead time ensures that the 1st RGC was preceded by a minimum period without activity (30ms) (**Figure 4A** [↗](#)).

We found that overall 2nd retinal spikes were more efficient in driving SC activity, and thus facilitating. This was true for both RGC-SC EXN (efficacy 1st RGC: median = 1.8 %, Q1 = 0.77 %, Q3 = 4.47 %; efficacy 2nd RGC: median = 2.66 %, Q1 = 1.16 %, Q3 = 6.07 %, p 1st vs 2nd = 3.13×10^{-25} , $n = 214$ pairs) and RGC-SC IN connected pairs (efficacy 1st RGC: median = 3.66 %, Q1 = 1.6, Q3 = 8.61; efficacy 2nd RGC: median = 4.33, Q1 = 2.03, Q3 = 8.25, p 1st vs 2nd = 1.68×10^{-6} , $n = 91$ pairs, Wilcoxon signed-rank test) (**Figure 4B** [↗](#), C). This effect is known as paired-spike enhancement and has been reported earlier in retinogeniculate (Mastronarde, 1987 [↗](#); Usrey et al., 1998 [↗](#)) and geniculocortical (Usrey et al., 2000 [↗](#)) connections. The majority of retinocollicular connected pairs showed paired-spike enhancement where 2nd retinal spikes were more effective in driving SC response (86.45% (185/214) RGC-SC EXN pairs, 80.22 % (73/91) RGC-SC IN pairs). The effect was more pronounced in RGC-SC EXN pairs where the 2nd retinal spike was 1.7 times as effective in driving responses in the postsynaptic SC neuron, while for RGC-SC IN pairs, the 2nd retinal spike was only 1.3 times more efficacious in driving SC spiking. To quantify this observation, we calculated the paired-spike ratio (PSR = efficacy 2nd/efficacy 1st) which confirmed a higher facilitation effect for RGCs connecting into excitatory neurons (PSR RGC-SC EXN = 1.711 ± 1.14 ; RGC-SC IN: 1.281 ± 0.414 , $p = 2.58 \times 10^{-7}$, Wilcoxon rank-sum, $n = 305$ pairs, **Figure 4D** [↗](#)).

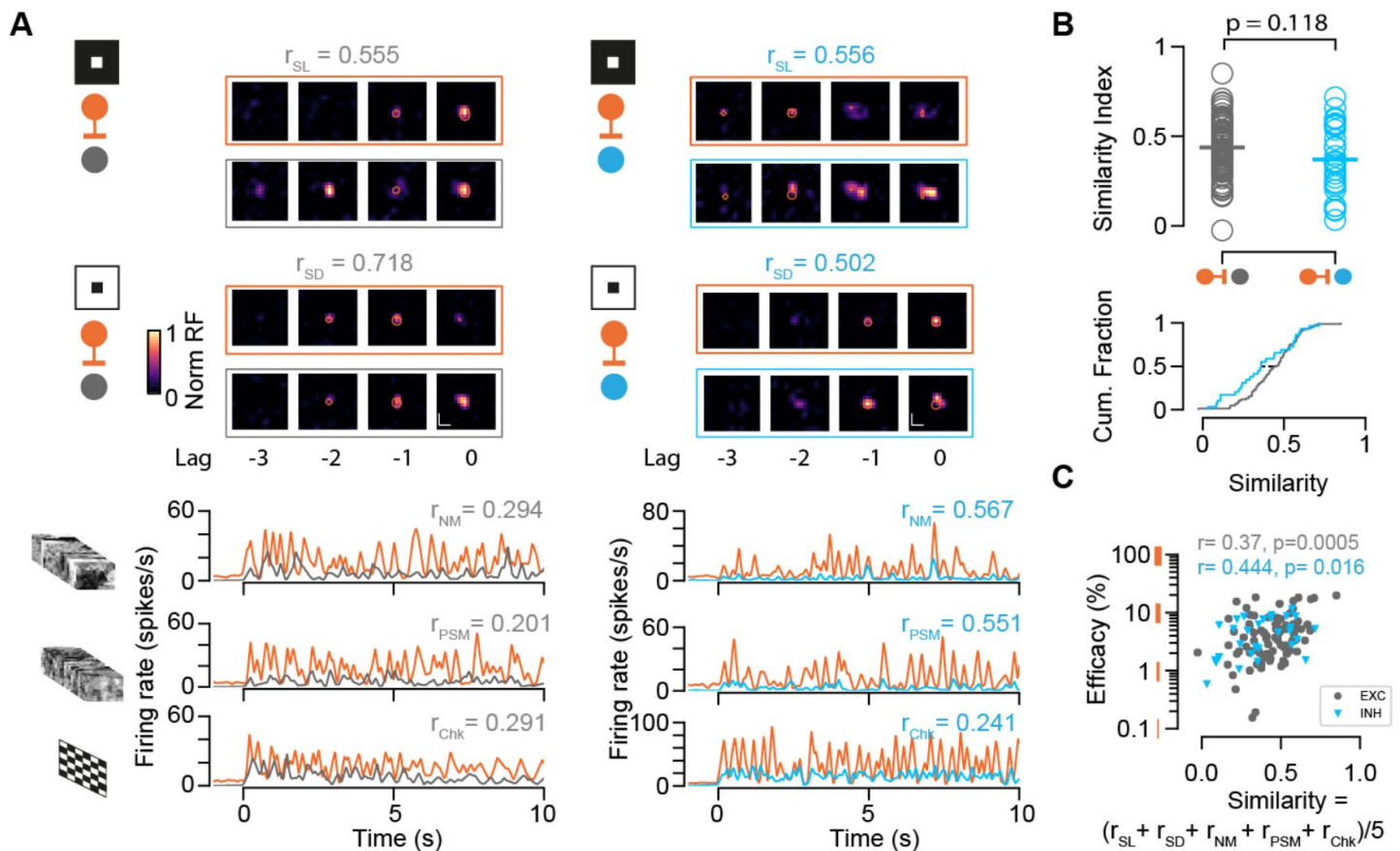


Figure 3

Characterization of functional similarity between retinocollicular connected pairs.

A, Top: Spatiotemporal receptive fields (STRF) evoked by dark (SD) and light (SL) sparse noise stimuli for RGC-SC EXN and RGC-SC IN connected pairs. Bottom: Visually evoked activity in response to a natural movie (NM), phase scrambled movie (PSM) and dense noise (Chk) stimulus. The functional similarity of the RGC axon and the postsynaptic SC neuron is characterized by the correlation coefficient during the different stimuli conditions (r_{SD} , r_{SL} , r_{NM} , r_{PSM} , r_{Chk}). **B**, The overall functional similarity between the presynaptic RGC and the postsynaptic SC neurons is reflected in the similarity index calculated from the averaged correlation coefficients $(r_{SL} + r_{SD} + r_{NM} + r_{PSM} + r_{Chk})/5$ ($p = 0.118$, two-sided Wilcoxon rank-sum test, $n = 85$ RGC-SC EXN pairs, $n = 29$ RGC-SC IN pairs). Scatter plot (top) and cumulative distribution (bottom) of similarity index. **C**, Relationship between similarity index and connection efficacy (Pearson correlation coefficient; $n = 85$ RGC-SC EXN pairs, $n = 29$ RGC-SC IN pairs).

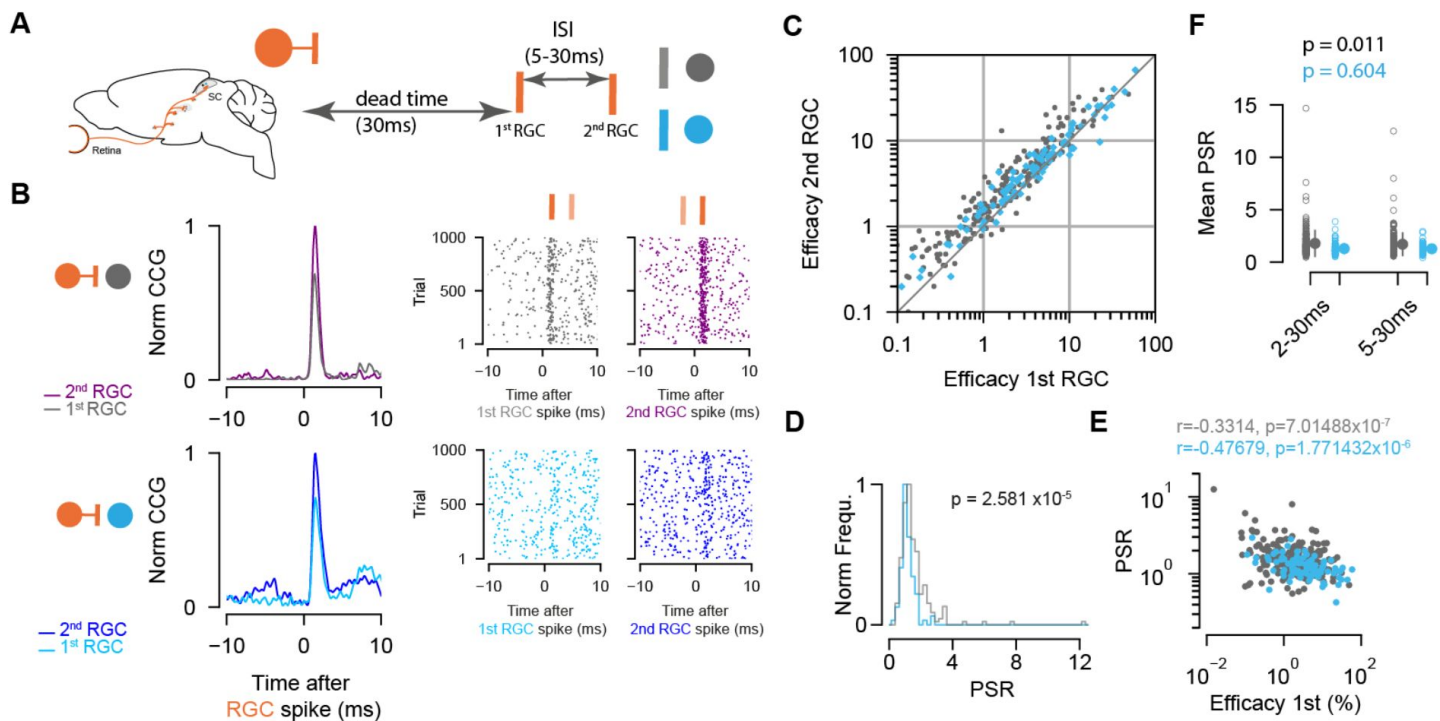


Figure 4

Paired-spike dynamics: Second retinal spikes are more efficient in driving SC response.

A, Schematic illustrating the temporal dynamics between two successive RGC spikes. Pairs of RGC spikes with a minimum inter-spike interval (ISI) of 5 ms and maximum ISI of 30 ms were included if there were no preceding spikes before the 1st RGC for a dead time of at least 30ms. **B**, Cross-correlograms (CCG) of example RGC-SC EXN and RGC-SC IN pairs calculated from spike trains selected for 1st and 2nd RGCs and the corresponding raster plots to 1000 trials triggered on RGC spikes. **C**, Scatter plot of efficacies for 1st versus 2nd retinal spikes. The majority of connected pairs (185/214 RGC-SC EXN pairs; 73/91 RGC-SC IN pairs) showed paired-spike enhancement in response to 2nd RGC spikes. **D**, Distribution of paired-spike ratio (PSR) for RGC-SC EXN and RGC-SC IN connected pairs. The paired pulse enhancement is stronger in SC-EXNs ($n = 214$ RGC-SC EXN; $n = 91$ RGC-SC IN, $p = 2.581 \times 10^{-5}$, two-sided Wilcoxon rank-sum test). **E**, Correlation of 1st RGC efficacy with the PSR for connected pairs ($r = 0.3314$ for $n = 214$ RGC-SC EXN pairs and $r = 0.4767$ for $n = 91$ RGC-SC IN pairs; logarithmic values; Pearson correlation coefficient test). **F**, Change in mean PSR depending on ISI for different groups of ISI intervals (2-3 ms and 5-30ms). The 2-30 ms group ensures to include high-firing RGCs (RGC-SC EXN pairs 2-30 vs. 5-30 ms $p = 0.011$; RGC-SC IN pairs 2-30 vs. 5-30 ms $p = 0.604$; Wilcoxon signed-rank test).

Previous work in the thalamocortical system has shown that weaker synaptic connections can be more strongly modulated compared to stronger synaptic connections (Ferrarese et al., 2018). To test the notion that also in the retinocollicular system, connected pairs with lower efficacy in response to 1st RGCs show greater enhancement in response to the 2nd RGCs, we correlated the efficacy of the 1st RGC to the PSR (**Figure 4E**). We observed a negative correlation between the logarithm of the 1st RGC efficacy and the logarithm of PSR for both RGC-SC EXN ($r = 0.3314$, $p = 7.0148 \times 10^{-7}$, $n = 214$ pairs) and RGC-SC IN pairs ($r = 0.4767$, $p = 1.7714 \times 10^{-6}$, $n = 91$ pairs). However, we found pairs that show low efficacy in response to the 1st RGC and low PSR. Taken together, we observed paired-spike enhancement in RGCs connecting onto both inhibitory and excitatory SC neurons, with a stronger facilitation effect for retinal afferents connecting onto excitatory, in comparison to inhibitory, SC target neurons. To estimate whether the ISI interval has an effect on the PSR, we added another group of 2-30 ms ISI interval to include RGCs that fire at high frequencies (PSR RGC-SC EXN = 1.783 ± 1.23 ; RGC-SC IN: 1.295 ± 0.528 , $p = 7.03 \times 10^{-6}$, Wilcoxon rank-sum, $n = 305$ pairs, **Figure 4F**). However, we did not observe a clear difference between the 2-30 and 5-30 ms groups for inhibitory connections (SC IN: $p = 0.604$, Wilcoxon signed-rank test). In contrast, the two different ISI groups of excitatory neurons show a statistically significant difference ($p = 0.011$, Wilcoxon signed-rank test), however, the effect size is small (Cohen's d RGC-SC EXN = 0.063, RGC-SC IN = 0.030).

Inhibitory SC neurons are less coupled to the retina compared to excitatory SC neurons

Our data so far reveals that excitatory and inhibitory SC neurons receive similarly strong drive from their retinal afferents. However, we observed a higher mean firing rate in SC INs compared to SC EXNs, on average (**Figure 1G**), which is intriguing given the similarity in the RGC drive on both cell types. A possible explanation could be that the activity of SC INs is less coupled to the RGC inputs as compared to that of SC EXNs, i.e. SC INs spikes are less often driven by their RGC inputs. To examine how strongly the activity of the two populations of SC neurons is coupled to the activity of individual RGC inputs, we estimated the synaptic coupling strength from the connection contribution (Levick et al., 1972; Sibille et al., 2022a; Usrey et al., 1999) of the measured retinocollicular pairs. The connection contribution reflects the fraction of postsynaptic SC spikes that are preceded by a presynaptic retinal spike within a short time window (**Figure 5A**). High contribution values indicate that SC neurons are primarily driven by individual RGC afferent inputs, while low contribution values point towards an integration of multiple inputs.

Interestingly, we found that the contribution of individual retinal spikes on SC activity was higher for excitatory compared to inhibitory SC neurons (**Figure 5B**) (contribution RGC-SC EXN: median = 18.27% Q1 = 10.79%, Q3 = 29.69%, $n = 214$; contribution RGC-SC IN: median = 12.2%, Q1 = 6.61%, Q3 = 23.02%, $n = 91$ connected pairs, $p = 0.0003$, two-sided Wilcoxon rank-sum test). This opens up the possibility that SC EXNs might be more specifically linked to the retinal input compared to SC INs. To assess the relationship between the coupling strength and the functional similarity, we correlated both measures and found a positive correlation, suggesting that stronger coupled pairs show higher similarity (RGC-SC EXN $r = 0.541$, $p = 8.9 \times 10^{-8}$, $n = 85$ pairs; RGC-SC IN $r = 0.461$, $p = 0.012$, $n = 29$ pairs, Pearson correlation coefficient test, **Figure 5C**). Moreover, we observed a diversity of firing rates in both SC EXNs and SC INs (**Figure 1G**) which suggests that low contribution values are related to the mean firing rates of the SC neurons. And indeed, when we correlated the contribution to the mean firing rate, we found a negative correlation for both RGC-SC EXN and RGC-SC IN pairs (RGC-SC EXN $r = -0.15079$, $p = 0.02742$; RGC-SC IN $r = -0.48386$, $p < 0.00000$, Pearson correlation coefficient test $n = 214$ RGC-EXN; $n = 91$ RGC-IN connected pairs) (**Figure 5D** and **Figure 5–figure supplement 1**).

Taken together, the cell-type specific difference in retinal coupling strength suggests that retinal input contributes a substantial part to the spiking activity of excitatory SC neurons, while inhibitory SC neurons seem to also integrate inputs from other sources. This finding denotes that

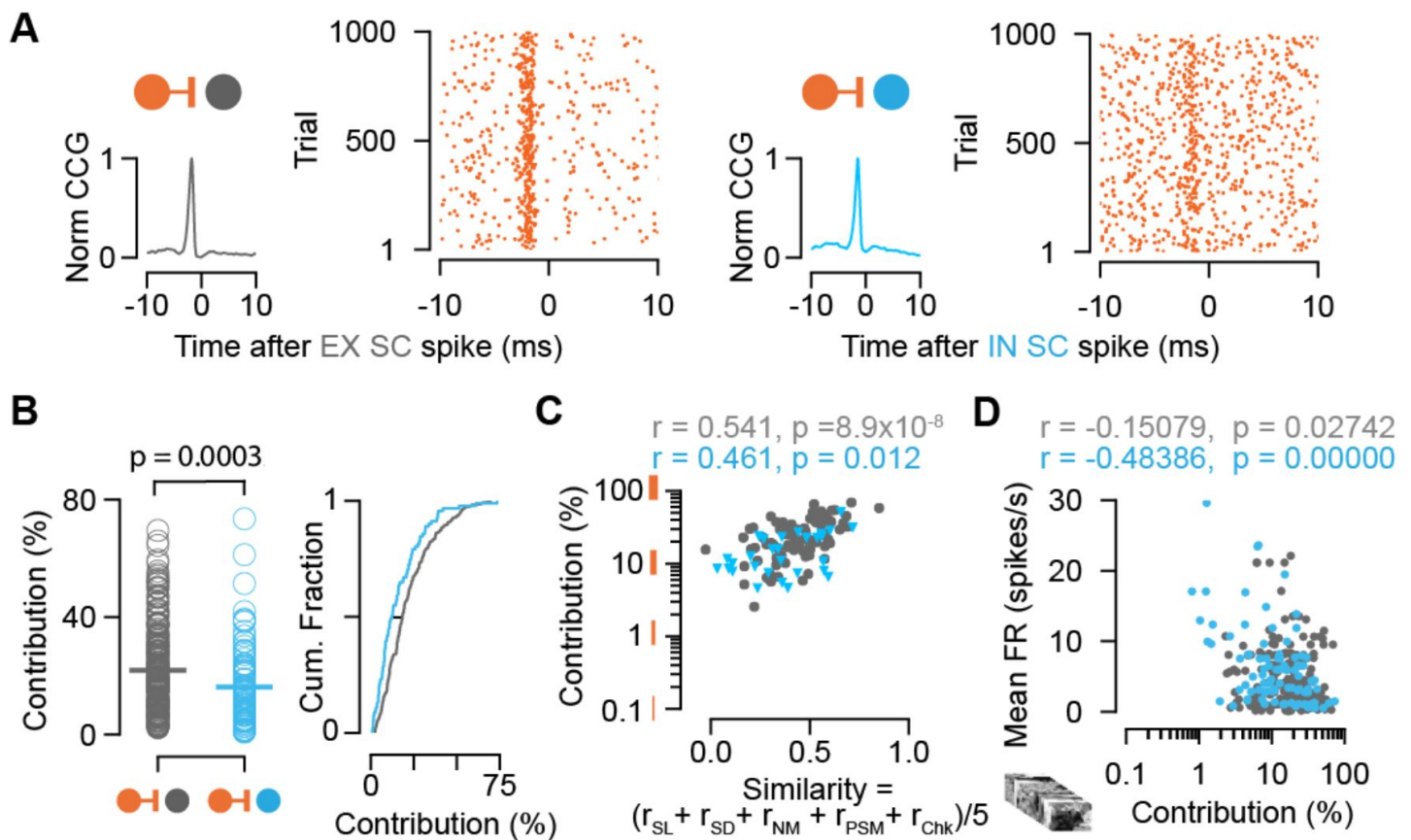


Figure 5

Connection contribution is higher in connected RGC-SC EXN pairs.

A, Example CCGs of monosynaptically connected RGC-SC EXN (top) and RGC-SC IN (bottom) pairs. Connected pairs were identified by their peaks in the CCGs (left). Raster plots of RGC spiking activity triggered on 1000 SC spikes (right). **B**, Scatter plot and cumulative distribution of contribution values for RGC-SC EXN and RGC-SC IN pairs ($n = 114$ RGC-SC EXN pairs, $n = 91$ RGC-SC IN pairs, $p = 0.0003$, two-sided Wilcoxon rank-sum test). **C**, Relationship between functional similarity index and connection contribution (Pearson correlation coefficient test; $n = 85$ RGC-SC EXN pairs, $n = 29$ RGC-SC IN pairs). **D**, Correlation of mean firing rate (to natural movie stimulus) and contribution for RGC-SC EXN and RGC-SC IN pairs (Pearson correlation coefficient, two-sided Wilcoxon rank-sum test, $n = 214$ RGC-EXN and $n = 91$ RGC-IN connected pairs; $n = 11$ penetrations from 9 mice).

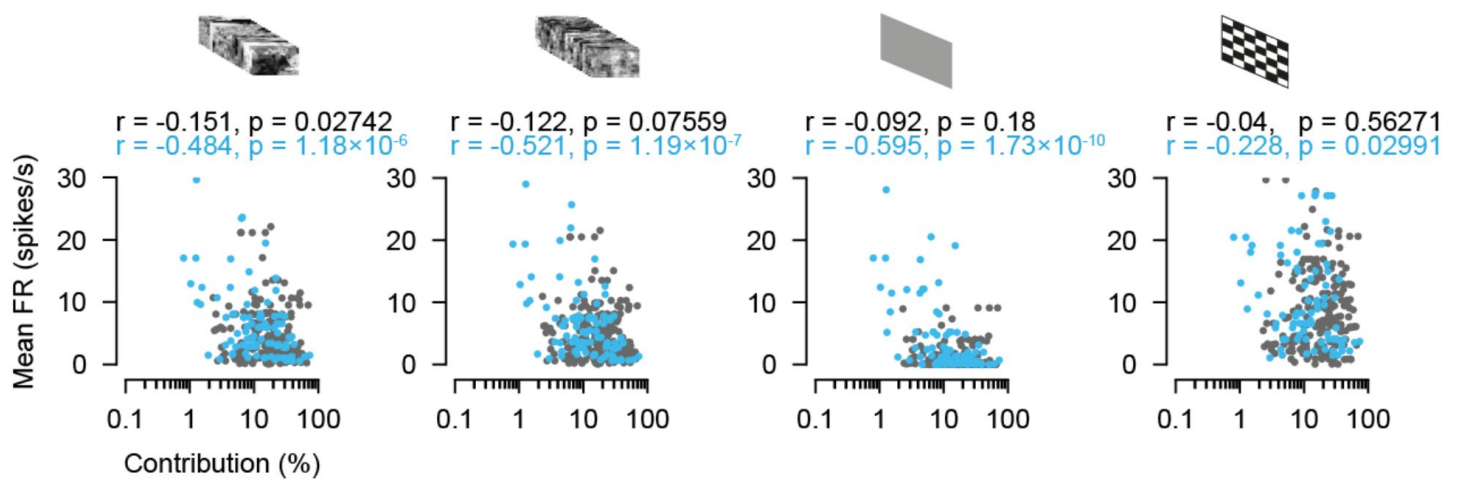


Figure 5

figure supplement 1 Contribution as a function of firing rate during different stimulus conditions.

Relationships between contribution and the conditions during natural movie (NM), phase scrambled movie (PSM), gray background (Bkg) and checkerboard (Chk) for RGC SC EXN and RGC-SC IN pairs RGC-SC EXN $n = 214$ pairs, RGC-SC INH $n = 91$ pairs, two-sided Wilcoxon rank-sum test).

Discussion

In this study, we elucidate how inhibitory and excitatory SC neurons integrate visual information from the retina *in vivo*. By combining the tangential Neuropixels recording approach with optotagging techniques, we were able to measure RGC-SC connected pairs in a cell-type specific manner and study their functional organization. Our findings allow the following conclusions: (1) The retinal drive is similarly strong for RGCs connecting onto GABAergic and non-GABAergic SC neurons. (2) The connection strength between the RGCs and both SC cell classes is positively correlated with their functional similarity, while still allowing for the presence of strong connections with low functional similarity. (3) Paired-spike enhancement in RGC-SC connections was a signature for both excitatory and inhibitory SC neurons. Finally, (4) the contribution of a retinal spike to overall SC firing is higher in excitatory SC neurons compared to inhibitory SC neurons.

We studied retinocollicular connectivity strength and aimed to answer the question whether excitatory and inhibitory SC populations use similar or different coding schemes to integrate visual information compared to the thalamocortical visual system. Connectivity patterns and circuit motifs have been extensively studied in the thalamocortical system, yet our understanding of the encoding mechanisms that govern retinocollicular signaling remains rudimentary albeit recent studies suggest that feedforward inhibition might be at play (Villalobos et al., 2018; Whyland et al., 2020). While cortical inhibitory neurons are more strongly activated by thalamic axons compared to excitatory neurons (Bruno and Simons, 2002; Cruikshank et al., 2007; Ji et al., 2016), our results now show that in the SC, the afferent retinal drive is comparably strong on excitatory and inhibitory neurons. This finding is intriguing and suggests that the two major visual brain structures in the vertebrate brain, the visual cortex and the SC, integrate afferent sensory information differentially.

Several parameters for example the degree of convergence (Kara and Reid, 2003), response properties (Alonso et al., 2001), synaptic mechanisms (Cruikshank et al., 2007) or spiking patterns (Usrey et al., 1998) can have meaningful effects on modulating the input strength on the target cell. SC neurons receive input from a small number (4-6) of RGCs (Chandrasekaran et al., 2007; Furman et al., 2013). In contrast, cortical neurons receive numerous (approximately one order of magnitude more) thalamic inputs that are usually weak (Bruno and Sakmann, 2006; Jin et al., 2011; Lien and Scanziani, 2018). Overall, we observed strong retinal drive onto both SC populations indicated by high efficacy values (**Figure 2**), and thus individual inputs may be sufficiently strong to drive the SC neuron. This is in contrast to thalamocortical inputs, where individually efficacy values are low, but the thalamic drive is boosted by synchronously active thalamic inputs (Alonso et al., 1996; Bruno and Sakmann, 2006). Our recording method allows for the simultaneous recording of multiple RGC afferents from SC neurons making it possible to evaluate how the synchrony among RGC afferents modulates the firing of postsynaptic SC neurons in future studies.

The difference in connection strength between cortical excitatory and inhibitory neurons could be partly attributed to their differences in response properties. Cortical inhibitory neurons display higher sensitivity, receive more convergent thalamic input, have higher firing rates and sample more randomly while excitatory neurons receive highly functional specific thalamic input (Kyriazi et al., 1994; Reid and Alonso, 1995; Simons and Carvell, 1989). While we observed higher firing rates in inhibitory SC neurons (**Figure 1G**), the efficacy of the RGC-SC connections was comparable between both types of neurons (**Figure 2F**), suggesting that the sensitivity of both neuron types is not markedly disparate. In general, visual response properties between excitatory

and inhibitory neurons in the SC may not differ to such a great extent (Inayat et al., 2015; Kasai and Isa, 2016; Shi et al., 2017), (but see (Barchini et al., 2018; Li and Meister, 2022)), and experiments in the inferior colliculus show that GABAergic and glutamatergic cells share similar response properties (Ono et al., 2017). These observations imply that in the SC and other parts of the midbrain, the cell-type specific difference in terms of selectivity might not be as strongly separated as in the cortex. Future studies are required to deeper investigate the differences in response properties of the subclasses of SC neurons and whether these properties are correlated to the RGC-SC connection strength. The positive correlation between efficacy and overall mean firing rate (**Figure 2G**), is intriguing. It suggests that highly excited SC neurons may transmit RGC inputs more efficaciously. Interestingly, Jouhanneau et al. (2018) reported a negative correlation between the distance of the excitatory postsynaptic potential peak to the action potential threshold and the efficacy (synaptic gain) in cortical PV neurons (Jouhanneau et al., 2018). Unfortunately, due to the inherent limitations of our extracellular recording approach, we are unable to investigate this aspect in our dataset. These findings underscore the complexity of synaptic transmission and wiring specificity and the multiple factors by which it can be influenced.

In our results, we observed an overall trend that functionally similar RGC-SC pairs show stronger connections which was true for both excitatory and inhibitory pairs (**Figure 3**). However, we also found pairs with considerably strong retinocollicular connectivity (efficacy values >10%) albeit the functional similarity between these connected cells was low (**Figure 3C**). This observation suggests that SC neurons may either show high projection-specificity by integrating inputs from a functional homogeneous population of RGCs (Li and Meister, 2022; Shi et al., 2017; Sibille et al., 2022a) or receive diverse inputs from multiple functional RGC types (Li and Meister, 2022; Sibille et al., 2022a). Moreover, a high degree of projection-specificity was recently shown to be maintained even disynaptically towards brain regions downstream of the SC (Reinhard et al., 2019), suggesting that at least one important role of the RGC-SC projection is to preserve the representation of the functionally diverse retinal channels (Baden et al., 2016). In our study, we investigated the functional similarity between connected RGC-SC pairs in response to sparse and dense noise stimuli as well as natural movie stimuli. And while we observed a diversity among the similarities between RGCs and connected targets in SC, supporting the idea that SC neurons may follow diverse wiring motifs (Sibille et al., 2022a), we observed no clear difference between inhibitory and excitatory SC neurons (**Figure 3B**). To further address the influence of functional properties on wiring mechanisms, it would be interesting to target the pool of distinct RGC types and to study how strongly they recruit SC subpopulations and modulate postsynaptic responses of their targets.

Temporal dynamics of presynaptic activity patterns are known to play a role in modulating synaptic strength (Usrey et al., 1998; Zucker, 1989). To study how the temporal spiking pattern of the RGC activity shapes RGC-SC connectivity, we analyzed the paired-spike dynamics. Across the population, we found a facilitating effect where responses to 2nd retinal spikes were more efficient in driving SC activity for both SC neuron populations. However, the paired-spike enhancement was more pronounced for excitatory than inhibitory neurons (**Figure 4**). Moreover, our observation highlights differences in the encoding strategies used by different regions of visual processing. While retinogeniculate connections show paired-spike enhancement (Usrey et al., 1998), thalamocortical and corticotectal synapses express mostly depressing dynamics (Bereshpolova et al., 2006; Swadlow et al., 2002; Swadlow and Gusev, 2001), but see (Usrey et al., 2000). Our data now reveals that retinocollicular connections facilitate, suggesting that the temporal integration of RGC inputs by SC neurons may be similar to the visual thalamus (dorsal lateral geniculate nucleus “dLGN”). The dLGN serves as another main target of RGC axons, however, in the mouse, the dLGN mainly consists of relay neurons, and only approximately 6% of inhibitory neurons (Butler, 2008; Evangelio et al., 2018). This disparity in interneuron density makes it challenging to directly compare the SC and dLGN circuits. Despite this difference, there exists a common synaptic motif between the dLGN and the SC. RGC axons innervating the dLGN

form triadic arrangements with excitatory and inhibitory neurons (Sherman, 2004 [↗](#); Wilson et al., 1984 [↗](#)) and this arrangement can also be found in the SC (Whyland et al., 2020 [↗](#)). It has been shown that GAD67 is a suitable marker to target the inhibitory neurons involved in the triadic motif in the SC (Whyland et al., 2020 [↗](#)) and therefore, it would be interesting to study retinal connectivity in the context of these triads in the SC using GAD67 mice.

In our study, we used cross-correlograms that were calculated from spike trains throughout the full recording period, to be applied on different connectivity strength measures. However, paired-spike dynamics are known to be modulated by the local temporal structure of inputs, for example the visual stimulus presented (Usrey et al., 1998 [↗](#)). In addition, facilitating and depressing effects can change with the target population (Bereshpolova et al., 2006 [↗](#)). Further studies are required to investigate whether the facilitating effect observed in our study changes upon targeting defined collicular subpopulations that may have differences in firing rates or during different stimulus conditions (e.g. spontaneous and stimulus evoked). In this study, retinocollicular connectivity was measured in anesthetized conditions. For future studies it would be interesting to examine connectivity dynamics and innervation strength in awake conditions as it has been shown that retinocollicular synapses are modulated by the animals' level of arousal (Schröder et al., 2020 [↗](#)). Likewise, more than 30 different types of RGCs have been characterized (Baden et al., 2016 [↗](#); Sanes and Masland, 2015 [↗](#)) and understanding how specific RGC types relay information onto the different SC neuron types would be important to address in more detail in future studies. Moreover, previous studies have shown a relationship between the response properties of SC neurons and their location along the dorsal/ ventral axis (Liu et al., 2023 [↗](#)) and differences in the density and size of retinocollicular terminals along this axis (Carter et al., 1991 [↗](#)). Exploring this aspect further utilizing a vertical recording approach could provide valuable insights in the context of functional organization.

Despite our observation that retinal drive is similarly strong onto both SC cell types, the contribution of retinal activity to individual SC neurons firing was higher in excitatory SC neurons. This opens up the possibility that excitatory SC neurons may be more specifically linked to retinal inputs, while inhibitory SC cells might receive excitation from other sources aside from the retina. This finding raises the question on what other sources could drive spiking in inhibitory SC neurons?

1. RGCs provide the main input to the superficial SC (Ellis et al., 2016 [↗](#)) with limited convergence (Sibille et al., 2022a [↗](#)). Nonetheless, the convergence may be higher for RGCs connecting onto inhibitory SC neurons which could explain the lower contribution and higher mean firing rates we found in inhibitory neurons. However, it has to be considered that our *in vivo* recording technique is limited in capturing the total pool of converging RGC inputs onto a neuron (Sibille et al., 2022a [↗](#)). Moreover, it has been shown that different types of RGCs exhibit variations in their axonal diameter, and this diameter is associated with how these cells relay information (Perge et al., 2009 [↗](#)). As a result, our recording approach may have a tendency to capture signals from RGCs with high firing rates that are located in close proximity to the Neuropixels probe. Viral retrograde transsynaptic tracing methods (Muellner and Roska, 2023 [↗](#); Rompani et al., 2017 [↗](#)), could be employed to study the full presynaptic RGC pool of single SC neurons.
2. In addition to retinal inputs, the SC receives extensive projections from the visual cortex (V1) that could act as a potential candidate to excite inhibitory neurons. And while indeed GABAergic horizontal cells have been shown to receive V1 input (Zingg et al., 2017 [↗](#)), the majority (93%) of corticotectal terminals target the dendrites of non-GABAergic neurons in the SC (Masterson et al., 2019 [↗](#)). Therefore, more work is needed and studying V1 excitation to defined inhibitory SC subpopulations will be an important topic to be investigated by the use of tracing studies and will shed light into this complexity.

3. Intracollicular connections can provide a source of recurrent excitation on SC neurons, amplifying the retinal drive (Shi et al., 2017 [↗](#)). However, the cell-type specificity of the intracollicular amplification is still debated. For example, while GAD2-positive horizontal cells receive excitatory inputs from stellate cells (Gale and Murphy, 2018 [↗](#)), intracollicular connectivity originating from neighboring excitatory cell types such as narrow-field (NF) and wide-field (WF) cells has shown to be rare (Gale and Murphy, 2018 [↗](#)). Next-generation *in vitro* multi-neuron patch-clamp recordings (Campagnola et al., 2022 [↗](#)) could reveal important new insights into the cell-type specific (Ayupe et al., 2023 [↗](#)) intracollicular wiring.
4. In addition, collicular neurons in the visual layers also get inputs from deeper SC layers. However, excitatory connections from deep SC to the upper layers are rather uncommon (Gale and Murphy, 2018 [↗](#); Ghitani et al., 2014 [↗](#)). Despite these studies have highlighted different sources of excitation onto inhibitory SC neurons, further studies are required that elucidate the role and strength of input that deep SC layers or external sources provide to drive inhibitory SC populations.

In this study, we recorded populations of well-isolated single units including RGCs as well as GABAergic and non-GABAergic SC neurons in VGAT-ChR2 mice *in vivo*. Our approach using VGAT-ChR2 mice to target all types of inhibitory SC neurons that express the vesicular GABA transporter VGAT enables a first impression on how inhibitory neurons sculpt retinocollicular signaling. Our inhibitory population is made up of 30% of the recorded SC neurons and connected pairs. These observations are in line with previous findings showing that in the upper SC layers, a substantial part of neurons (30%) are GABAergic (Ranney Mize, 1992 [↗](#); Whyland et al., 2020 [↗](#)) and that around one third of the postsynaptic targets of retinocollicular terminals are GABAergic (Boka et al., 2006 [↗](#); Whyland et al., 2020 [↗](#)).

In the last decade, the diversity of SC cell types has been mainly characterized based on morphology and response properties (Franceschi and Solomon, 2018 [↗](#); Ito et al., 2017 [↗](#)). Morphologically and functionally, four distinct classes of retinorecipient SC neurons have been identified: stellate, horizontal, wide-field and narrow field neurons (Gale and Murphy, 2016 [↗](#), 2014 [↗](#)). Excitatory SC cell types such as wide-field, stellate and narrow-field cells differ in morphological characteristics. This morphological diversity might also be linked to how these cells integrate inputs from the retina. Smaller cells, such as narrow-field (NF) and stellate cells, typically have smaller RF sizes. Their limited dendritic arborization enables them to selectively respond to specific visual features. NF neurons show strong direction selectivity (DS) suggesting that these cells might be strongly driven by the retina, likely from DS-RGCs (Gale and Murphy, 2014 [↗](#)). In contrast, wide-field cells have extensive dendritic arbors and large RF sizes. These cells likely integrate inputs from multiple RGC axons (Gale and Murphy, 2014 [↗](#)).

The use of genetic mouse lines provides a valuable tool for understanding the circuitry and functional roles of different neuron classes by selectively targeting specific cell types. Advanced techniques have started to shed light on this diversity by providing cell-type characterizations on the molecular level (Barchini et al., 2018 [↗](#); Gale and Murphy, 2018 [↗](#), 2014 [↗](#); Li and Meister, 2022 [↗](#); Villalobos et al., 2018 [↗](#)). Three categories of potential GABAergic interneurons have been classified in the SC by the use of Cre-lines (Whyland et al., 2020 [↗](#)) that can be further subdivided into intrinsic interneurons and potential projection neurons. A recent study by Tsai et al. (2022) [↗](#) has mapped retinocollicular connectivity by targeting specific RGC-SC circuits using a novel approach (Trans-Seq) to classify subsets of SC cell-types that are innervated by genetically defined RGC types (Tsai et al., 2022 [↗](#)). They identified five inhibitory clusters of retinorecipient SC cells that allow to specifically map retinocollicular connectivity. In addition, 12 distinct inhibitory subtypes across the entire SC circuit have been recently identified using single-nucleus RNA sequencing (Ayupe et al., 2023 [↗](#)). Thus, we are at an enthralling stage where information on SC cell-types finally becomes available based on gene expression patterns, alike to the cortical circuit (Tremblay et al., 2016 [↗](#)). Likewise investigating how optogenetic manipulation of SC INs

modulates functional properties of recorded SC EXNs (Gale and Murphy, 2016 [↗](#)) is crucial for gaining a mechanistic understanding of visual processing within the SC circuitry. Hence, a key step forward would be to leverage the tracing and transcriptomics studies and combine these methods with our high-density recording approach to enable a finer characterization of the collicular interneuron diversity and the complexity of their connectivity patterns *in vivo*.

Taken together, our study targeted a heterogeneous population of GABAergic neurons in VGAT-ChR2 mice and our results reveal that excitatory and inhibitory neurons in the two main brain areas for visual processing, the visual cortex and the superior colliculus, integrate sensory afferent inputs in different ways.

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Author contributions

C.G. and J.K. conceived and designed the study; C.G. collected the data; C.G., J.S., and J.K. analyzed the data and C.G. and J.K. wrote the manuscript with inputs from J.S.

Declaration of interests

The authors declare no competing interests.

Materials and methods

Experimental Design

Animals

All experiments were carried out in accordance with the guidelines given by Landesamt für Gesundheit und Soziales (LAGeSo - G 0142/18) Berlin and were approved by the authority. Adult VGAT-ChR2 (Slc32a1-COP4*H134R/EYFP, The Jackson Laboratory, stock no. 014548, n= 9) transgenic mice of either sex were used to activate inhibitory neurons (Zhao et al., 2011 [↗](#)). Mice were aged 7-9 months old on the day of recordings.

Surgical procedures

All experiments were conducted in VGAT-ChR2 mice of both sexes. Mice were induced with 2.0-2.5% isoflurane (CP-Pharma) in an induction chamber and transferred to the surgery. Once anesthetized, the surgery was performed in a stereotactic frame (Narishige) with a closed-loop temperature controller (FHC DC) for monitoring the animal's body temperature and maintaining it at 37°C. The isoflurane level was gradually lowered during surgery and maintained at 0.8-1.5% while ensuring a complete absence of vibrissa twitching or responses to tactile stimulation. During surgery, the eyes were protected with eye ointment (Vidisic, Bausch+Lomb). Mice were head-fixed

in the frame, and the skull was exposed. The head was aligned along the anterior-posterior axis and marks were made for the craniotomy at 600-1500 μm ML and 0-1500 μm AP from Lambda (Paxinos and Franklin, Nixdorf 2007 stereotactic atlas) using a micromanipulator (Luigs and Neumann). For optotagging experiments, an additional mark for the optic fiber placement was made at 3500 μm AP, 500-1200 μm ML to Bregma. A headpost was placed on the skull and implanted with dental cement (Paladur). Dental cement was also used to build a recording chamber to provide a bath for grounding. During this step, a silver wire (AG-10W, Science Products) was attached to the dental cement chamber for grounding. It is important to keep the dental cement chamber low enough on the anterior and posterior sides to allow for the shallow angle probe and fiber implantations, see Sibille et al (2022) for a detailed description of the method (Sibille et al., 2022b [DOI](#)). Craniotomies were made at the marked positions using a dental drill (Johnson-Promident). To allow for the probe insertion at a horizontal angle between 15° and 20°, a small part of the skull at the posterior part of the craniotomy was slightly thinned using the drill to ensure a smooth insertion (Sibille et al., 2022b [DOI](#)).

Anesthetized extracellular recordings and optogenetic stimulation

Electrophysiological recordings in SC were performed in 9 anesthetized mice of both sexes (4 female, 5 male) using Neuropixels 1.0 probes (Jun et al., 2017 [DOI](#)). Data were acquired at a sampling rate of 30 kHz using the Open Ephys acquisition software and GUI (Siegle et al., 2017 [DOI](#)) (www.open-ephys.org) (<http://www.open-ephys.org/>) [DOI](#) and the PXIe hardware acquisition system (National Instrument NI-PXIe-1071). The signal was amplified and stored in both the local field potential band (LFP, high pass-filtered 0-300 Hz) and the action potential band (AP, 300 Hz to 3 kHz). The tangential recordings along the anterior-posterior axis in SC were performed as described in Sibille et al. (2022) (Sibille et al., 2022a [DOI](#)). The fiber for optogenetic stimulation was inserted in the SC prior to the Neuropixels probe implantation to decrease mechanical pressure and reduce drift of the electrode probe. The angled optogenetic fiber was zeroed at the brain surface and inserted from the anterior in a 60-70° angle towards the AP axis (see **Figure 1A** [DOI](#)) and slowly lowered using a manual micromanipulator until reaching an insertion of 150-200 μm . Following the fiber implantation, the Neuropixels probe was inserted tangentially in a 15 to 20° angle in the superior colliculus (SC) using a micromanipulator (NewScale). The probe was zeroed at Lambda and inserted 500-1200 μm ML and 2800-3800 μm DV. After the Neuropixels probe was implanted, the optic fiber was lowered further slowly (to a maximum insertion depth of ~250 μm) and LED pulses were applied to test for optogenetic activation of cells. After correct placement, the probe and fiber were allowed to settle for 15-20 minutes before data acquisition was started. In addition, confirmation of probe placement within the SC was done via receptive field (RF) mapping to check the number of channels with RFs. RFs were estimated from the multi-unit activity (MUA) using spike-triggered average (STA) (Sibille et al., 2022b [DOI](#)) (**Figure 1C** [DOI](#)).

Optotagging to identify GABAergic neurons in VGAT-ChR2 mice

We identified (“optotagged”) populations of single GABAergic SC neurons via Channelrhodopsin-2 (ChR2) activation in VGAT-ChR2 mice. The optic fiber was calibrated prior to each experiment to ensure a stable light output (2.5mW) measured at the fiber tip. Before the electrode probe insertion (see above), the optic fiber (NA .66, 200 μm core diameter, with a bare fiber tip, PlexBright, Plexon) coupled to a blue (470 nm) LED module (PlexBright, Plexon) was inserted using a 60-70° angle tilted towards the AP axis (**Figure 1A** [DOI](#)). Blue light pulses were used to evoke spiking activity in VGAT-expressing neurons and to verify the proper alignment of the optic fiber and the Neuropixels probe. If no optogenetically evoked neuronal activity was observed, either the fiber was lowered to a maximum of 250 μm or the probe position was re-adjusted to ensure both the activation and recording of GABAergic SC neurons. Once the Neuropixels electrode probe and optic fiber were properly aligned, optogenetic stimulation using either a square wave or Gaussian function stimulus pulse was applied to identify light-activated channels (**Figure 1A** [DOI](#), B). As the stimulation using a square wave pulse led to the induction of light artefacts in the spiking response (Jun et al., 2017 [DOI](#)), we manually removed these artefacts in a post-processing step by

interpolating the raw values in each channel during the onset and offset of the square pulse in 2 out of 11 experiments. In order to decrease light artefacts induced by the sharp light onset in the square wave pulse, we changed the pulse to using a Gaussian function stimulus pulse to gradually increase light intensity in the remaining 8 experiments. The light pulses were applied during spontaneous conditions (presentation of a black background, 100 ms pulse duration, 2 Hz, 200 trials, 2.7–3.5 mW measured at the fiber tip) and conditions with visually evoked activity (checkerboard stimulus, 1000 ms pulse duration, 30 trials ON, 30 trials OFF, 2.7–3.5 mW measured at the fiber tip, $n = 10/11$ experiments). Both stimulation protocols were shown twice within the recording to ensure stable responses throughout the recording session. GABAergic neurons were identified by their increased short-latency responses to blue light stimulation pulses. When we compared the visually evoked responses to those induced by ChR2 activation we found that they varied in their temporal activation profile (**Figure 1B**, C). The response to the visual sparse noise stimulus starts ~80ms after the stimulus onset (**Figure 1C**) while responses induced by ChR2 activation had a shorter latency following light onset (**Figure 1C**).

Visual stimulation

Visual stimuli were generated in Python using the PsychoPy2 toolbox (Peirce et al., 2019). The visual stimuli were displayed on a calibrated screen (Dell, refresh rate = 120 Hz, mean luminance = 120 cd/m²). For receptive field mapping, we presented the sparse noise visual stimulus on a grid of 24x 14 squares. The sparse noise targets were either dark (on light background) or light (on dark background) and 10 deg in size with [n_targets per frame, n_trials per position] = [2,20] and presented for 100 ms. A set of standardized different visual stimuli was presented including a natural movie (30 trials, randomized with other stimuli, 10s) taken from Froudarakis et al. (2014) and a full-field chirp stimulus (Baden et al., 2016). The frametimes of the visual stimuli were marked by stimulus-locked synchronizing events (TTLs).

Histology

For histological reconstruction of the electrode track, the probe was removed, coated with fluorescent dye (DiI, Abcam-ab145311) and re-inserted in the same location. In a few experiments we performed multiple recordings and only stained the last insertion. Subsequently, the animal was terminated with an excess of isoflurane (> 4%). Cardiac perfusion was performed with phosphate buffer saline (PBS) followed by 4% paraformaldehyde (PFA) in PBS. Brains were post-fixed overnight in 4% PFA, and stored in PBS until slicing (100 μ m) on a vibratome (Leica VT1000S). The brain slices were mounted using a mounting medium containing DAPI (DAPI-Fluoromount-G, Biozol Cat. 0100-20). Slices were imaged on a fluorescence microscope using a 2.5x objective for post hoc visualization of the electrode probe track.

Data analysis and statistical analyses

Data analysis was performed in Python 3.0 (www.anaconda.com (<http://www.anaconda.com/>)) using custom-written scripts. MATLAB 2019b (www.mathworks.com (<http://www.mathworks.com/>)) was used for spike sorting, see below. Statistical tests were performed using the two-sided Wilcoxon rank-sum test for unpaired samples and the Wilcoxon signed-rank test (two-sided) for paired samples using the scipy.stats module in Python, unless stated otherwise. For correlation analyses we used Pearson correlation implemented via the linregress function. Population results are indicated as mean $\pm$ standard deviation if not stated otherwise. The number of neurons and animals used in each analysis is reported in the Results section or figure legends.

Multi-unit activity extraction

For MUA analysis, common average referencing was applied to the bandpass filtered AP data (Butterworth filter order 2, 0.3 to 3 kHz), where each event is extracted using custom-written Python scripts. Spike detection was performed for each channel independently at a threshold of 4

standard deviations in the AP band (double side detection). The scripts for performing the MUA analysis are available on the GitHub repository (https://github.com/KremkowLab/tangential_recording)

Offline Spike sorting and criteria for unit classification

KiloSort2 and 3 (Pachitariu et al., 2016) (<https://github.com/MouseLand/Kilosort>) were used for automatic detection and clustering of spikes followed by manual curation using phy2 (Rossant et al., 2016) (<https://github.com/cortex-lab/phy>). Double-counted spikes were removed for each cluster (within ± 0.16 ms) (Siegle et al., 2021). Furthermore, clusters with between-unit overlapping spikes that show above chance zero-lag peaks in the cross-correlograms (CCG, peak windows ± 0.5 ms) were re-evaluated individually in phy to be either refined or removed. Inter-spike-interval (ISI) violations were calculated as the ratio of the spikes within the refractory period (± 1.5 ms) to the total number of spikes. Units with ISI $> 0.05\%$ were removed. We also checked for stable firing of clusters throughout the recording. Furthermore, we excluded units from further analysis that were lying outside the optogenetic stimulation range (see below, Identification of Inhibitory neurons, see **Figure 1B**) and outside the SC boundaries (from STA RFs on MUA, see **Figure 1C**).

Waveform classification to identify RGC axons and SC neurons

A waveform classification approach was applied to distinguish action potentials (APs) from retinal ganglion cell (RGC) axons and somatic APs from SC neurons (Sibille et al., 2022a). In brief, we calculated the multi-channel waveform (MCW) which reflects the spatiotemporal profile of the AP signals. Afferent axons and somatic signals could be classified based on their distinct waveform which allowed us to classify clusters into afferent axon AP vs somatic AP. In essence, due to the SC electrode probe implantation along the anterior-posterior (AP) axis, the electrode captures signals from retinal axons that innervate the SC along the electrode channels. This propagation of RGC axonal action potentials along the axonal path is visible in the spatiotemporal profile of their multi-channel waveforms (**Figure 1D**, left).

Waveform classification approach to separate GABAergic and non-GABAergic SC neurons

We performed standard waveform analyses to attempt to separate GABAergic SC neurons based on waveform features (Jia et al., 2019; Quirk et al., 2009) (**Figure 1 – figure supplement 2**). Briefly, we detected the negative (trough) and positive peak (peak) from single-channel waveforms and calculated their peak-to-trough duration. The amplitude was defined as the absolute difference between trough and peak. The peak-to-trough ratio was defined as the ratio between amplitudes of peak and trough.

Post hoc separation of GABAergic and non-GABAergic SC neurons

To study retinal input integration in a cell-type specific manner, we used a method to separate the neural responses of GABAergic neurons from that of excitatory neurons in the SC. First, we identified the minimum and maximum recording sites on the probe that were photo-activated. These borders were defined via the minimum and maximum peak channel of light-induced units that are activated on the probe. SC neurons that did not fall inside this optogenetic stimulation range (identified by their peak channel) were excluded from further analysis. We employed stringent criteria for determining the boundaries of LED evoked response and selection of neurons to be included (**Figure 1 – figure supplement 1**). The distribution of excitatory and inhibitory SC neurons is not significantly different for 9/11 experiments (Wilcoxon rank-sum, p values = 0.307, 0.0115, 0.755, 0.834, 5.01×10^{-6} , 0.79, 0.80, 0.26, 0.33, 0.08, 0.13). In the two significantly different experiments only 2 RGC-SC EXN pairs were located in the region without identified SC INs, and thus will not affect the results. SC neurons were separated based on their

short-latency increase in responses to blue light pulses during two different protocols: (1) baseline activity (black background, 100 ms pulse duration, 2 Hz pulse frequency) and (2) during a checkerboard stimulus (1000 ms pulse duration, 30 trials LED ON, 30 trials LED OFF). This classification was done using a custom-written graphical-user-interface (GUI) approach where units were labeled as either “light-responsive”, “non light-responsive”. The labeling step was conducted by three independent observers. A unit was included if 2 out of 3 observers chose the same label. Units with unstable firing throughout the recording period were excluded from further analysis.

Identification of synaptically connected RGC-SC pairs

Monosynaptic connections between RGC axons and SC neurons were detected using established methods (Bereshpolova et al., 2020 [↗](#); Clay Reid and Alonso, 1995 [↗](#); Denman and Contreras, 2015 [↗](#); Usrey et al., 1998 [↗](#)) based on statistically significant peaks at synaptic delays. Briefly, we calculated a jittered (timescale jitter 5 ms) version of each spike train by randomizing all spike times within a consecutive 10-15 ms window (Smith and Kohn, 2008 [↗](#)). We then calculated the cross-correlation between a pair of neurons both for the original (raw CCG) and the jittered spike train (jittered CCG). Subtracting the jittered CCG from the raw CCG results in a jitter-corrected version of the CCG which was used for monosynaptic pairs detection. Connected pairs were identified using cross-correlation analysis and significant peaks (peak detection window from + 0.5 to 3.5 ms) had to extend over the threshold (baseline + 4x standard deviation) for at least 5 consecutive time bins (0.1 ms resolution) (as described previously, in Sibille et al. (2022) (Sibille et al., 2022a [↗](#))). The cross-correlations were calculated using the Python pycorrelate package (<https://github.com/tritemio/pycorrelate> [↗](#)). Please note that for CCG analysis, spike trains were correlated over the full recording duration, except for periods with optogenetic stimulation. Latencies to peak response were calculated from the CCGs as the maximum time within the response window.

Analysis of retinocollicular connection strength

Synaptic efficacy and contribution measures of connected RGC-SC pairs were estimated from spike trains during the entire recording session (except for optotagging periods which were excluded) using standard approaches (Bereshpolova et al., 2020 [↗](#); Clay Reid and Alonso, 1995 [↗](#); Swadlow and Gusev, 2002 [↗](#); Usrey et al., 1998 [↗](#)). Briefly, efficacy was estimated from the baseline-corrected CCGs by dividing the area of the CCG peak (baseline window: > -1 to < 0.4 ms, peak window: > 0.4 to < 3.5 ms) by the total number of presynaptic spikes. Thus, an efficacy measure of 1 (100%) would reflect that for each presynaptic retinal spike, a postsynaptic spike could be detected. To estimate the connection contribution, we counted the number of SC spikes that were preceded by a retinal afferent spike, integrated over a time window between -3 to -0.5 ms, and divided this number by the total number of SC spikes. A contribution of 1 (100%) indicates that all spikes of an SC neuron are preceded by retinal afferent spikes.

Paired-spike dynamics analysis

For estimation of paired-spike dynamics, baseline-corrected CCGs were re-calculated triggered on 1st and 2nd RGCs using the same baseline and peak windows as mentioned above. We identified pairs of retinal spikes (1st and 2nd RGC) for each RGC-SC connected pair (Usrey et al., 1998 [↗](#)). Briefly, a pair of retinal spikes was defined by a specific inter-spike interval (ISI) where the second RGC spike followed the 1st RGC by a ISI of 5 to 30 ms. The second criterion for classifying a pair of retinal spikes was a certain dead time where the 1st retinal spike was preceded by a minimum period of silence (dead time: 30ms). The dead time criterion was applied to ensure a comparable level of activity immediately preceding all first spikes. The preceding window (> -1 to < 0.4 ms) was used as a baseline window. Paired-spike enhancement was calculated as the ratio of efficacy for the 2nd RGC to the efficacy for the 1st RGC.

Functional similarity analysis of retinocollicular connected pairs

To characterize the functional similarity between connected RGCs and the target neurons in the SC, we estimated the functional similarity index. To do so, we calculated the correlation coefficients from responses during the light and dark sparse noise (r_{SL} and r_{SD}) as well as during the natural movie stimulus (r_{NM}), phase scrambled movie (r_{PSM}) and during a checkerboard stimulus (r_{Chk}). The similarity index was calculated between the responses of the RGC and the responses of the postsynaptic SC neuron. As a result, the similarity index reflects the degree of similarity in the responses of the connected pairs. The correlation of activity during sparse noise stimuli (r_{SL} and r_{SD}) was estimated from the spatiotemporal receptive fields (STRFs) which were calculated from the STA. The similarity during the other stimuli (r_{NM} , r_{PSM} and r_{Chk}) were calculated from the PSTHs. We limited this analysis step to neurons with high signal-to-noise ratio (SNR) in the visually evoked activity ($SNR > 8$ for the STRFs). The overall similarity index was then calculated from the averaged correlation coefficients functional similarity index = $(r_{SD} + r_{SL} + r_{NM} + r_{PSM} + r_{Chk})/5$.

Data availability

The datasets that support the findings of this study are available from the corresponding author on request.

Code availability

Custom Python code is available from the corresponding author on request.

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Author information

Carolyn Gehr

Neuroscience Research Center, Charité-Universitätsmedizin Berlin, Charitéplatz 1, 10117 Berlin, Bernstein Center for Computational Neuroscience Berlin, Philippstraße 13, 10115 Berlin, Institute for Theoretical Biology, Humboldt-Universität zu Berlin, Philippstraße 13, 10115 Berlin, Einstein Center for Neurosciences Berlin, Charitéplatz 1, 10117 Berlin
ORCID iD: [0000-0003-0892-506X](https://orcid.org/0000-0003-0892-506X)

Jérémie Sibille

Neuroscience Research Center, Charité-Universitätsmedizin Berlin, Charitéplatz 1, 10117 Berlin, Bernstein Center for Computational Neuroscience Berlin, Philippstraße 13, 10115 Berlin, Institute for Theoretical Biology, Humboldt-Universität zu Berlin, Philippstraße 13, 10115 Berlin, Einstein Center for Neurosciences Berlin, Charitéplatz 1, 10117 Berlin

Jens Kremkow

Neuroscience Research Center, Charité-Universitätsmedizin Berlin, Charitéplatz 1, 10117 Berlin, Bernstein Center for Computational Neuroscience Berlin, Philippstraße 13, 10115 Berlin, Institute for Theoretical Biology, Humboldt-Universität zu Berlin, Philippstraße 13, 10115 Berlin, Einstein Center for Neurosciences Berlin, Charitéplatz 1, 10117 Berlin
ORCID iD: [0000-0001-7077-4528](https://orcid.org/0000-0001-7077-4528)

Editors

Reviewing Editor

Marla Feller

University of California, Berkeley, United States of America

Senior Editor

Sacha Nelson

Brandeis University, United States of America

Reviewer #1 (Public Review):

Gehr and colleagues used an elegant method, using neuropixels probes, to study retinal input integration by mouse superior collicular cells in vivo. Compared to a previous report of the same group, they opto-tagged inhibitory neurons and defined the differential integration onto each group. Through these experiments, the author concluded that overall, there is no clear difference between the retina connectivity to excitatory and inhibitory superior colliculus neurons. The exception to that rule is that excitatory neurons might be driven slightly stronger than inhibitory ones. Technically, this work is performed at a high level, and the plots are beautifully conceived, but I have doubts if the interpretation given by the authors is solid. I will elaborate below.

Some thoughts about the interpretation of the results.

My main concern is the "survivor bias" of this work, which can lead to skewed conclusions. From the data set acquired, 305 connections were measured, 1/3 inhibitory and 2/3 excitatory. These connections arise from 83 RGC onto 124 RGC (I'm interpreting the axis of Fig.2 C). Here it is worth mentioning that different RGC types have different axonal diameters (Perge et al., 2009). Here the diameter is also related to the way cells relay information (max frequencies, for example). It is possible that thicker axons are easier to measure, given the larger potential changes would likely occur, and thus, selectively being picked up by the neuropixels probe. If this is the case, we would have a clear case of "survival bias", which should be tested and discussed. One way to determine if the response properties of axonal termini are from an unbiased sample is to make a rough functional characterization as generally performed (see Baden et al. 2006). This is fundamental since all other conclusions are based on unbiased sampling.

One aspect that is not clear to me is to measure of connectivity strength in Figure 2. Here it seems that connectivity strength is directly correlated with the baseline firing rate of the SC neuron (see example plots). If this is a general case, the synaptic strength can be assumed but would only differ in strength due to the excitability of the postsynaptic cell. This should be tested by plotting the correlation coefficient analysis against the baseline firing rate.

My third concern is the assessment of functional similarity in Fig. 3. It is not clear to me why the similarity value was taken by the arithmetic mean. For example, even if the responses are identical for one connected pair that exclusively responds either to the ON or OFF sparse noise, the maximal value can only be 0.67. Perhaps I misunderstood something. Secondly, correlations in natural(istic) movies can differ dramatically depending on the frame rate that the movie was acquired and the way it is displayed to the animal. What looks natural to us will elicit several artifacts at a retinal level, e.g., due to big jumps between frames (no direction-selective response) or overall little modulation (large spatial correlations). I would rather opt for uniform stimuli, as suggested previously. Of course, these are also approximations but can be easily reproduced by different labs and are not subjected to the intricacies of the detailed naturalistic stimulus used.

Fourth. It is important to control the proportion of inhibitory cells activated optogenetically across the recording probe. Currently, it is not possible to assess if there are false negatives. One way of controlling for this would be to show that the number of inhibitory interneurons doesn't vary across the probe.

Fifth. In Fig. 4, the ISI had a minimal bound of 5 ms. Why? This would cap the firing rate at 200Hz, but we know that RGC in explants can fire at higher frequencies for evoked responses. I would set a lower bound since it should come naturally from the after-depolarization block. Another aspect that remains unclear is to what extent the paired-spike ratio depends on the baseline firing rate. This would change the interpretation from the particular synaptic connection to the intrinsic properties of the cell and is plausible since the baseline firing rate

varies tremendously. One related analysis would be to plot the change of PSR depending on the ISI. It would be intuitive to make a scatter plot for all paired spikes of all recorded neurons (separated into inhibitory and excitatory) of ISI vs. PSR.

Panel 4E is confusing to me. Here what is plotted is efficacy 1st against PSR (which is efficacy 2nd/efficacy 1st). Given that you have a linear relation between efficacy 1st and efficacy 2nd (panel 4C), you are essentially re-plotting the same information, which should necessarily have a hyperbolic relationship: $f(x) = y/x$. Thus, fitting this with a linear function makes no sense and it has to be decaying if efficacy 2nd > efficacy 1st as shown in 4C.

Finally, in Figure 5, the perspective is inverted, and the spike correlations are seen from the perspective of SC neurons. Here it would also be good to plot the cumulative histograms and not look at the averages. Regarding the similarity index and use of natural stats, please see my previous comments. Also, would it be possible to plot the contribution v/s the firing rate with the baseline firing rate with no stimulation or full-field stimulation? This is important since naturalistic movies have too many correlations and dependencies that make this plot difficult to interpret.

Overall, the paper only speaks from excitatory and inhibitory differences in the introduction and results. However, it is known that there are three clear morphologically distinct classes of excitatory neurons (wide-field, narrow-field, and stellate). This topic is touched in the discussion but not directly in the context of these results. Smaller cells might likely be driven much stronger. Wide-field cells would likely not be driven by one RGC input only and will probably integrate from many more cells than 6.

Reviewer #2 (Public Review):

This study follows up on a previous study by the group (Sibille et al Nature Communications 2022) in which high density Neuropixel probes were inserted tangentially through the superficial layers of the superior colliculus (SC) to record the activity of retinocollicular axons and postsynaptic collicular neurons in anesthetized mice. By correlating spike patterns, connected pairs could be identified which allowed the authors to demonstrate that functionally similar retinal axon-SC neuron pairs were strongly connected.

In the current study, the authors use similar techniques in vGAT-ChR2 mice and add a fiber optic to identify light-activated GABAergic and non-light-activated nonGABAergic neurons. Using their previously verified techniques to identify connected pairs, within regions of optogenetic activation they identified 214 connected pairs of retinal axons and nonGABAergic neurons and 91 pairs of connected retinal axons and GABAergic neurons. The main conclusion is that retinal activity contributed more to the activity of postsynaptic nonGABAergic SC neurons than to the activity of postsynaptic GABAergic SC neurons.

The study is very well done. The figures are well laid out and clearly establish the conclusions. My main comments are related to the comparison to other circuits and further questions that might be addressed in the SC.

It is stated several times that the superior colliculus and the visual cortex are the two major brain areas for visual processing and these areas are compared throughout the manuscript. However, since both the dorsal lateral geniculate nucleus (dLGN) and SC include similar synaptic motifs, including triadic arrangements of retinal boutons with GABAergic and nonGABAergic neurons, it might be more relevant to compare and contrast retinal convergence and other features in these structures.

The GABAergic and nonGABAergic neurons showed a wide range of firing rates. It might be interesting to sort the cells by firing rates to see if they exhibit different properties. For example, since the SC contains both GABAergic interneurons and projection neurons it would

be interesting to examine whether GABAergic neurons with higher firing rates exhibit narrower spikes, similar to cortical fast spiking interneurons. Similarly, it might be of interest to sort the neurons by their receptive field sizes since this is associated with different SC neuron types.

The recording techniques allowed for the identification of the distance between connected retinocollicular fibers and postsynaptic neurons. It might also be interesting to compare the properties of connected pairs recorded at dorsal versus ventral locations since neurons with different genetic identities and response properties are located in different dorsal/ventral locations (e.g. Liu et al. Neuron 2023). Also, regarding the strength of connections, previous electron microscopy studies have shown that the retinocollicular terminals differ in density and size in the dorsal/ventral dimension (e.g Carter et al JCN 1991).

Was optogenetic activation of GABAergic neurons ever paired with visual activation? It would be interesting to examine the receptive fields of the nonGABAergic neurons before and after activation of the GABAergic neurons (as in Gale and Murphy J Neurosci 2016).

Reviewer #3 (Public Review):

This study performs in vivo recordings of neurons in the mouse superior colliculus and their afferents from the retina, retinal ganglion cells (RGCs). Building on a preparation they previously published, this study adds the use of optogenetic identification of inhibitory neurons (aka optotagging) to compare RGC connectivity to excitatory and inhibitory neurons in SC. Using this approach, the authors characterize connection probability, strength, and response correlation between RGCs and their target neurons in SC, finding several differences from what is observed in the retina-thalamus-visual cortex pathway. As such, this may be a useful dataset for efforts to understand retinocollicular connectivity and computations.

Author Response

The following is the authors' response to the original reviews.

We are grateful for the comments from the reviewers, which helped us to strengthen our analyses and communicate more effectively the details of our findings and their significance. To address their criticisms, we have performed new analyses and revised the text and figures. We believe the manuscript was significantly improved. We provide the line number of important parts of the text that were changed, here in this letter. Below, we address the specific comments from the reviewers in detail.

Reviewer #1 (Public Review):

Gehr and colleagues used an elegant method, using neuropixels probes, to study retinal input integration by mouse superior collicular cells in vivo. Compared to a previous report of the same group, they opto-tagged inhibitory neurons and defined the differential integration onto each group. Through these experiments, the author concluded that overall, there is no clear difference between the retina connectivity to excitatory and inhibitory superior colliculus neurons. The exception to that rule is that excitatory neurons might be driven slightly stronger than inhibitory ones. Technically, this work is performed at a high level, and the plots are beautifully conceived, but I have doubts if the interpretation given by the authors is solid. I will elaborate below.

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My main concern is the "survivor bias" of this work, which can lead to skewed conclusions. From the data set acquired, 305 connections were measured, 1/3 inhibitory and 2/3 excitatory. These connections arise from 83 RGC onto 124 RGC (I'm interpreting the axis of Fig.2 C). Here it is worth mentioning that different RGC types have different axonal diameters (Perge et al., 2009). Here the diameter is also related to the way cells relay information (max frequencies, for example). It is possible that thicker axons are easier to measure, given the larger potential changes would likely occur, and thus, selectively being picked up by the neuropixels probe. If this is the case, we would have a clear case of "survival bias", which should be tested and discussed. One way to determine if the response properties of axonal termini are from an unbiased sample is to make a rough functional characterization as generally performed (see Baden et al. 2006). This is fundamental since all other conclusions are based on unbiased sampling.

First of all, we want to thank the reviewer for the detailed and constructive comments based on which we refined the analysis and updated the figures. We hope that our changes adequately address the concerns of reviewer #1.

We would like to clarify that Fig. 2C represents an example from a single experiment. In total, we recorded 326 RGCs and 680 SC neurons in total, with 161 individual RGCs making connections onto 183 individual SC neurons. Moreover, we thank the reviewer for bringing up that important point about the potential "survivor bias". To address this concern, we would like to provide some clarifications (see below). In addition, we now added the point that different RGCs can have different axonal diameter as requested by the reviewers (line 605).

It is important to note that our approach does not capture the total pool of retinal inputs. Moreover, we did not want to convey the impression that our approach equally captures all retinal inputs to a given SC neuron, as this is not the case. Likewise, it is important to note that our current method does not allow for the measurement of axonal diameters. To obtain an estimate of axonal thickness, complementary techniques such as imaging/staining or electron microscopy would be needed. Our study aimed at characterizing connected RGC-SC pairs and how excitatory and inhibitory neurons in the SC integrate retinal inputs, providing valuable insight on their wiring principles.

We greatly appreciate the reviewer for highlighting this limitation and we now address these points in the discussion of the revised version of our manuscript (line 603).

Regarding the suggested "rough functional characterization" of the RGCs. We have thought about this analysis and unfortunately, we did not present the necessary stimuli, e.g. chirp, in all experiments to be able to perform this analysis. Moreover, the dataset represented in this work contains only 326 RGC neurons, with 161 identified RGCs making connections to SC neurons. Thus, it is unlikely that our dataset uniformly covers all ~30 RGC types in the mouse. However, given that our dataset is the first measurement of RGC inputs to SC INs and SC EXNs in vivo, we believe it provides a first step and a foundation for future studies focusing on specific RGC types to refine our understanding of the RGC-SC circuitry. We discuss this point now in the revised manuscript (line 586).

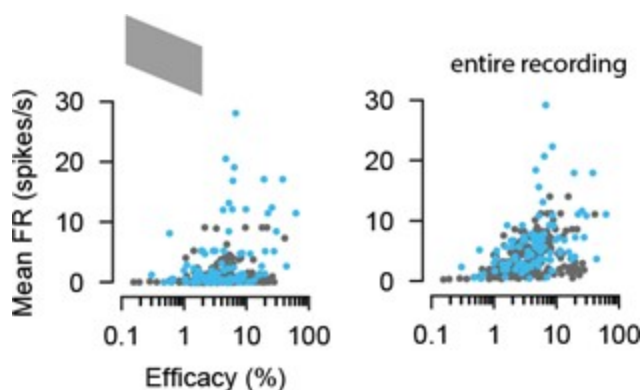
One aspect that is not clear to me is to measure of connectivity strength in Figure 2. Here it seems that connectivity strength is directly correlated with the baseline firing rate of the SC neuron (see example plots). If this is a general case, the synaptic strength can be assumed but would only differ in strength due to the excitability of the postsynaptic cell. This should be tested by plotting the correlation coefficient analysis against the baseline firing rate.

We appreciate the reviewer for bringing up this important point. From the analysis perspective, we would like to clarify that the efficacy measure is independent of the baseline firing rate. It quantifies the probability of adding spikes on top of the baseline rate by subtracting the baseline firing rate before measuring the area of the peak (Usrey et al., 1999).

Furthermore, we acknowledge the reviewer's interesting and valuable observation about the relationship of the firing rate and the excitability of the SC neuron in the example plots. To test whether the efficacy is directly related to the mean firing rate, we conducted additional analyses to show the efficacy measure as a function of the mean firing rate (Author response image 1 and Figure 2G). To that end, we utilized two different measures of firing rate: the mean firing rate during spontaneous activity (gray screen) over a duration of 10 sec (across 30 trials), which was interleaved with the natural movie presentations, and the overall firing rate throughout the entire recording session. Our findings indeed reveal a positive correlation, as predicted by the reviewer (Author response image 1, gray screen: EXC $r = 0.22721$; $p < 0.00081$; INH: $r = 0.34677$, $p = 0.00076$; entire recording: EXC $r = 0.42685$; $p < 0.0005$; INH: $r = 0.43543$, $p = 0.00002$).

Author response image 1.

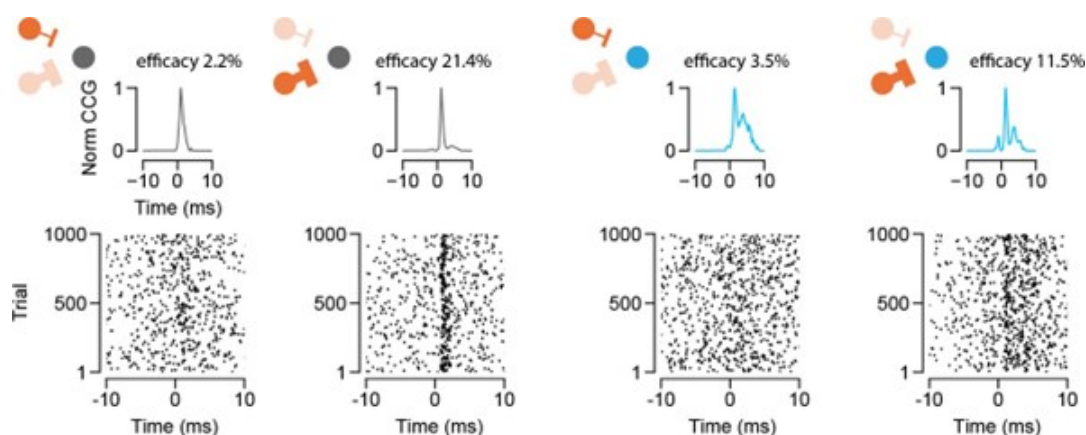
Efficacy measure of connected RGC-SC pairs as a function of the mean firing rate during different stimulus conditions: during spontaneous activity (gray screen, left) and throughout the entire recording session (right).



However, it is important to note that although we observe a correlation on the population level, the relationship between postsynaptic firing and efficacy is diverse. We identify pairs with strong connections despite the firing rate of the postsynaptic SC cell being low. Likewise, we also find pairs with weak connections despite the firing rate of the SC neuron being high (Author response image 2). These observations suggest that factors beyond the postsynaptic firing contribute to the efficacy of the connection. This is exemplified by the fact that SC neurons can receive both strong and weak connections from their convergent presynaptic RGC pool.

Author response image 2.

RGC-SC connectivity. Cross-correlograms showing 4 connected RGC-SC pairs (top) with two RGCs connecting onto the same SC neuron. Raster plots of SC neuron spiking activity in response to firing of the presynaptically connected RGC. The same SC neuron can receive both strong and weak RGC inputs.



In summary, we thank the reviewer for bringing up this important question, and we believe that our additional analyses shed light on the relationship between firing rate and efficacy. This result is very interesting, and we include these findings in the updated Figure 2 in the revised manuscript (panel 2G) in exchange with the panel of the peak latency. Moreover, we also address this point now in the results and discussion section of the revised manuscript (line 280 and line 525).

My third concern is the assessment of functional similarity in Fig. 3. It is not clear to me why the similarity value was taken by the arithmetic mean. For example, even if the responses are identical for one connected pair that exclusively responds either to the ON or OFF sparse noise, the maximal value can only be 0.67. Perhaps I misunderstood something.

We thank the reviewer for raising this point about the clarification regarding the calculation of the similarity index. We apologize for any confusion caused by our description on the similarity index calculation. To clarify, the similarity index was calculated specifically between the responses of the RGC and the responses of the postsynaptic SC neuron, rather than between the neurons and the visual stimulus. As a result, the similarity index reflects the degree of similarity in the responses of the connected pairs. Therefore, if the responses of the RGC and the connected postsynaptic neuron are identical, regardless of whether they respond exclusively to ON, only to OFF, or a mixture of ON-OFF, the similarity index will be one. We have updated the relevant part in the methods section to make this point clearer to the reader (line 917).

Secondly, correlations in natural(istic) movies can differ dramatically depending on the frame rate that the movie was acquired and the way it is displayed to the animal. What looks natural to us will elicit several artifacts at a retinal level, e.g., due to big jumps between frames (no direction-selective response) or overall little modulation (large spatial correlations). I would rather opt for uniform stimuli, as suggested previously. Of course, these are also approximations but can be easily reproduced by different labs and are not subjected to the intricacies of the detailed naturalistic stimulus used.

We agree with the reviewer that spatiotemporal correlations of naturalistic stimuli are complex. To address this point, we added two stimuli with little spatiotemporal correlations to the similarity analysis. The first stimulus we added is a phase scrambled version of the natural movie (PSM, also taken from Froudarakis et al. (2014)). The second is a binary white noise checkerboard stimulus. These stimuli were presented randomly interleaved with the natural movie, for 30 trials each. The similarity index analysis revealed that even with

uniform stimuli included, the average similarity index is correlated to the efficacy. We show this data now in Figure 3.

Fourth. It is important to control the proportion of inhibitory cells activated optogenetically across the recording probe. Currently, it is not possible to assess if there are false negatives. One way of controlling for this would be to show that the number of inhibitory interneurons doesn't vary across the probe.

We thank the reviewer for highlighting this important aspect of the experiment and analysis. We are aware of this point and therefore took extra care to minimize the biases that could be introduced by our recording and stimulation method. Our approach to include recorded excitatory and inhibitory neurons was conservative, briefly:

1. We included only excitatory and inhibitory neurons that were within the SC, defined by visually driven activity and continuous retinotopy (see method).
2. We further restricted the included neurons to neurons that were located within the boundaries of the LED evoked responses, i.e. the recording channels with optogenetic evoked MUA responses within the SC (Figure 1 – figure supplement 1).
3. Both excitatory and inhibitory SC neurons were selected in this way.

These inclusion criteria were specifically designed to avoid sampling excitatory neurons from regions on the Neuropixels probe that lacked optogenetically evoked responses and thus to minimize the number of falsely labeled excitatory neurons.

To illustrate these inclusion criteria and the resulting spatial distribution of the selected excitatory and inhibitory SC neurons along the 384 channels of the Neuropixels probe, we now added a supplementary figure (Figure 1 – figure supplement 1). This figure shows the multi-unit activity in response to optogenetic stimulation and the distribution of inhibitory and excitatory single units within the range of channels that are activated via LED stimulation for 3/11 selected experiments. This highlights that we employed stringent criteria for determining the boundaries and selecting which neurons to include in our study. The distribution of excitatory and inhibitory SC neurons is not significantly different for 9/11 experiments (Wilcoxon rank-sum test, p values = 0.307, 0.0115, 0.755, 0.834, 5.01 $\times 10^{-6}$, 0.79, 0.80, 0.26, 0.33, 0.08, 0.13). Moreover, in the two significantly different experiments only 2 RGC-SC EXC pairs were located in the region without identified SC INs, and thus will not affect the results. We now address this point in the methods section (line 859).

Fifth. In Fig. 4, the ISI had a minimal bound of 5 ms. Why? This would cap the firing rate at 200Hz, but we know that RGC in explants can fire at higher frequencies for evoked responses. I would set a lower bound since it should come naturally from the after-depolarization block.

The chosen 5 ms minimal bound was in the range used in previous literature, e.g. 4-30 ms in Usrey et al. 1998 (Usrey et al., 1998). To address the question of the reviewer, we re-analyzed the data with a lower bound of 2 ms (2 – 30 ms) to include RGCs that fire at higher frequencies than 200Hz. However, we did not observe a clear difference between the 2-30 and 5-30 ms groups for inhibitory connections (SC IN: $p = 0.604$). Only the excitatory connections show a statistically significant difference ($p = 0.011$), however, the effect size is small (Cohen's $d = \text{EXC} = 0.063$, $\text{INH} = 0.030$). Nonetheless we updated a panel in figure 4 to represent the 2-30 ms group (Figure 4F).

Another aspect that remains unclear is to what extent the paired-spike ratio depends on the baseline firing rate. This would change the interpretation from the particular

synaptic connection to the intrinsic properties of the cell and is plausible since the baseline firing rate varies tremendously.

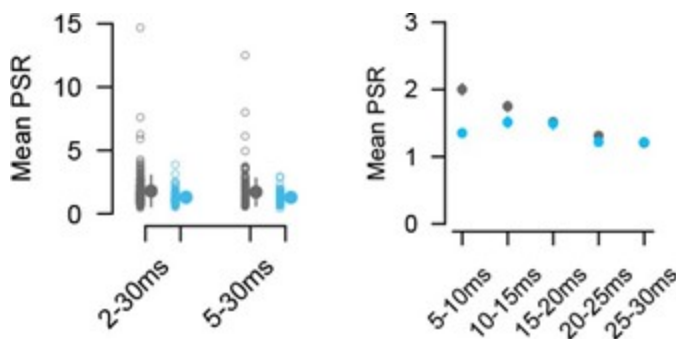
To address how the paired-spike ratio depends on the baseline firing rate we plotted the change of PSR depending on ISI as suggested by the reviewer.

One related analysis would be to plot the change of PSR depending on the ISI. It would be intuitive to make a scatter plot for all paired spikes of all recorded neurons (separated into inhibitory and excitatory) of ISI vs. PSR.

We appreciate the valuable suggestion from the reviewer. We have now separated the ISIs into distinct groups spanning 5 ms intervals represented in Author response image 3, right. These intervals range from 5-10 ms up to 25-30 ms. Notably, we observe a difference between the excitatory and inhibitory populations. The excitatory population exhibits a monotonic decrease in mean PSR across the intervals, while the inhibitory population shows a peak around 10/15 ms.

Author response image 3.

Change of mean paired-spike ratio (PSR) depending on ISI. Left) Comparison of PSR between two groups of different ISIs. The 2-30 ms group ensures to include high-firing RGCs (excitatory pairs 2-30 vs 5-30 ms $p = 0.011$; inhibitory pairs 2-30 vs 5-30 ms $p = 0.604$, Wilcoxon signed-rank). Right) PSR for groups of different ISI intervals. Mean PSR $\pm$ SEM for excitatory groups: 2.0 ± 0.09 , 1.75 ± 0.09 , 1.51 ± 0.05 , 1.31 ± 0.05 , 1.2 ± 0.05 ; inhibitory groups: 1.35 ± 0.06 , 1.51 ± 0.09 , 1.5 ± 0.1 , 1.22 ± 0.06 , 1.21 ± 0.07 . p E vs I (within group): 1.55×10^{-5} , 9.55×10^{-2} , 4.21×10^{-1} , 3.74×10^{-1} , 6.22×10^{-1} , Wilcoxon rank-sum test.



Panel 4E is confusing to me. Here what is plotted is efficacy 1st against PSR (which is efficacy 2nd/efficacy 1st). Given that you have a linear relation between efficacy 1st and efficacy 2nd (panel 4C), you are essentially re-plotting the same information, which should necessarily have a hyperbolic relationship: $[f(x) = y/x]$. Thus, fitting this with a linear function makes no sense and it has to be decaying if efficacy 2nd > efficacy 1st as shown in 4C.

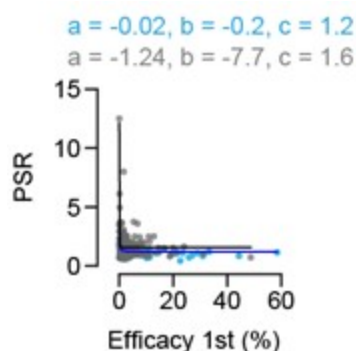
We thank the reviewer for raising this question which helped us to improve the representation and disruption of the results shown in figure 4. Panel 4E is intended to investigate whether there is a correlation between the efficacy strength (eff 1st) and the amount of facilitation (PSR). From panel 4C it is already evident that the data points for high efficacies lie closer to the unity line, as compared to the data points for low efficacies. This suggests that the PSR is stronger for connections with smaller efficacies 1st. To quantify this relationship, we have plotted the efficacy 1st vs the PSR in panel 4E, which thus adds new information to the figure. Importantly, this panel is shown in log-log scales, and therefore the

decaying relationship is not evident. If we had shown the data on linear-linear scale, the decaying function would have been evident (Author response image 4). And indeed, as the reviewer pointed out, we cannot fit a hyperbolic relationship with a linear function. This is exactly the reason why we show the data in log-log scale and also estimate the Pearson correlation also from the logs of the efficacies and PSRs.

In Author response image 4 we show the relationship plotted on linear scale using an approach to fit the hyperbolic relationship employing a hyperbolic cosecant function $a/\sinh(b * x) + c$.

Author response image 4.

Relationship between efficacy to 1st RGC and PSR visualized on linear scale using a hyperbolic fitting approach $a/\sinh(b * x) + c$.



Finally, in Figure 5, the perspective is inverted, and the spike correlations are seen from the perspective of SC neurons. Here it would also be good to plot the cumulative histograms and not look at the averages.

We added the cumulative histogram in Figure 5 (panel B), in addition to represent the raw data points and the mean.

Regarding the similarity index and use of natural stats, please see my previous comments. Also, would it be possible to plot the contribution v/s the firing rate with the baseline firing rate with no stimulation or full-field stimulation? This is important since naturalistic movies have too many correlations and dependencies that make this plot difficult to interpret.

We now show the contribution vs firing rates for different stimulus conditions in a new figure supplement (Figure 5- figure supplement 1). We added the correlations to the different stimuli for baseline firing rate with no stimulation (gray background), full-field stimulation (checkerboard) and phase scrambled natural movie.

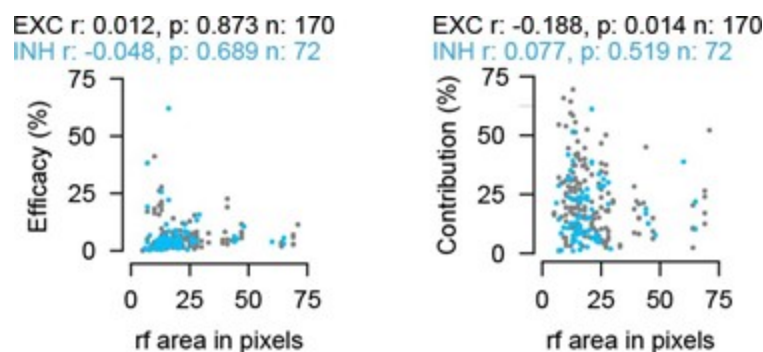
Overall, the paper only speaks from excitatory and inhibitory differences in the introduction and results. However, it is known that there are three clear morphologically distinct classes of excitatory neurons (wide-field, narrow-field, and stellate). This topic is touched in the discussion but not directly in the context of these results. Smaller cells might likely be driven much stronger. Wide-field cells would likely not be driven by one RGC input only and will probably integrate from many more cells than 6.

We thank the reviewer for this comment. We agree with the reviewer that addressing how the different excitatory and inhibitory cell-types integrate RGC input is important to understand the visual processing mechanisms in the SC. The presented study aimed at comparing the excitatory and inhibitory population in general using the VGAT-ChR2 mouse line. Understanding how specific genetically defined cell-types integrate RGC inputs is clearly very interesting and should be done. Unfortunately, the mouse lines that would allow targeting genetically identified inhibitory cell-types are still limited and therefore we can only use functional measurements to assess different types of neurons in the SC. We now address this point about distinct SC cell-types in the discussion (line 643).

One possible functional measurement is the size of the receptive field, which, to some degree, could be used as a proxy for different morphologies, i.e. small receptive fields could hint towards compact morphology while large receptive fields could indicate a wider morphology. It is known for example that narrow-field and stellate cells have small RF sizes, while wide-field cells have large RFs. We studied the relationship between the RF size and spike waveform duration but did not find a significant correlation (Figure R6). Moreover, the spike waveform duration, as discussed in the manuscript, is not a valid criterion to separate EXNs and INs in the SC, as it is common practice in the cortex. We now also looked into whether the connectivity strength is related to the RF size. Interestingly, while in the current dataset we do not find a significant correlation between the efficacy and the receptive field size for both EXN and IN (Author response image 5, left), we do find a significant negative correlation between contribution and receptive field size for the excitatory neurons (Author response image5, right). This result indicates that SC excitatory neurons with small receptive fields are more strongly coupled to the RGC input as compared to neurons with larger receptive fields.

Author response image 5.

Relationship between RF size and connectivity measures (efficacy and contribution) for RGC-SC EXN and RGC-SC IN pairs (two-sided Wilcoxon rank-sum test).



Reviewer #2 (Public Review):

This study follows up on a previous study by the group (Sibille et al Nature Communications 2022) in which high density Neuropixel probes were inserted tangentially through the superficial layers of the superior colliculus (SC) to record the activity of retinocollicular axons and postsynaptic collicular neurons in anesthetized mice. By correlating spike patterns, connected pairs could be identified which allowed the authors to demonstrate that functionally similar retinal axon-SC neuron pairs were strongly connected.

In the current study, the authors use similar techniques in vGAT-ChR2 mice and add a fiber optic to identify light-activated GABAergic and non-light-activated nonGABAergic neurons. Using their previously verified techniques to identify connected pairs, within regions of optogenetic activation they identified 214 connected pairs of retinal axons and nonGABAergic neurons and 91 pairs of connected retinal axons and GABAergic neurons. The main conclusion is that retinal activity contributed more to the activity of postsynaptic nonGABAergic SC neurons than to the activity of postsynaptic GABAergic SC neurons.

The study is very well done. The figures are well laid out and clearly establish the conclusions. My main comments are related to the comparison to other circuits and further questions that might be addressed in the SC.

It is stated several times that the superior colliculus and the visual cortex are the two major brain areas for visual processing and these areas are compared throughout the manuscript. However, since both the dorsal lateral geniculate nucleus (dLGN) and SC include similar synaptic motifs, including triadic arrangements of retinal boutons with GABAergic and nonGABAergic neurons, it might be more relevant to compare and contrast retinal convergence and other features in these structures.

Thank you for pointing out that crucial point. Indeed, the comparison to the thalamus is a valid argument, as both the SC and LGN are primary targets of RGC axon terminals. During the preparation of the manuscript, we extensively discussed whether to compare our new SC dataset with existing literature on the LGN or the primary visual cortex (V1) is the more appropriate. Ultimately, we decided on using the visual cortex as the main comparison because of the following reasons:

1. The SC is widely recognized as an evolutionary conserved circuit for visual computation and visually guided behaviors, while the dLGN is generally regarded as a relay station for RGC information to the visual cortex (Steriade, McCormick, 1997). Thus, we believe it is more relevant to compare the evolutionary older visual circuit (SC) to the evolutionary newer visual circuit (visual cortex).
2. In the mouse, the dLGN contains only a limited number of inhibitory interneurons and represent only approximately 6% of the total dLGN neuronal population (Butler, 2008; Evangelio et al., 2018). It has been suggested that the rodent somatosensory thalamus even lacks interneurons (Arcelli et al., 1997). Consequently, directly comparing inhibitory interneurons in the SC to those in the dLGN would pose challenges.
3. Along the same line, the density and also the diversity of inhibitory neurons in the SC is high and likely more comparable to the density and diversity of inhibitory neurons in the visual cortex, than to the dLGN circuit. In the dLGN, TC projection neurons far outnumber inhibitory neurons (Arcelli et al., 1997; Evangelio et al., 2018) and the dLGN is inhabited by just 1-2 classes of GABAergic retinorecipient interneurons (Arcelli et al., 1997; Jaubert-Miazza et al., 2005; Krahe et al., 2011; Ling et al., 2012). Classification approaches (e.g. 3D reconstruction) so far have not revealed any subclasses except for distinctions in intrinsic membrane properties (Leist et al., 2016), suggesting low interneuron diversity in the dLGN. This is in contrast to the vLGN, where a recent study found a diversity of GABAergic neurons (Sabbagh et al., 2021).
4. In the thalamo-cortical circuit, there exists a notable difference in how cortical excitatory and cortical inhibitory neurons are driven by their thalamic input (Alonso and Swadlow, 2005; Cruikshank et al., 2007). This discrepancy forms the basis for several models of visual processing in the visual cortex (Kremkow et al., 2016; Taylor

That said, the reviewer is correct that the dLGN and the SC share certain wiring motifs, such as the triadic arrangements of retinal boutons. Unfortunately, the VGAT-ChR2 mouse line used in our study does not specifically label SC inhibitory neurons that are involved in the formation of triadic arrangements. Therefore, we are unable to draw specific conclusion regarding this point. To further investigate this aspect, the usage of GAD67 mice, which have been shown to selectively label intrinsic interneurons which receive RGC input and contact non-GABAergic dendrites (Whyland et al., 2020), would be necessary. Nonetheless, we acknowledge the question raised by the reviewer and in response, we have now provided a more in-depth comparison to the dLGN in the discussion section of the revised manuscript (line 565).

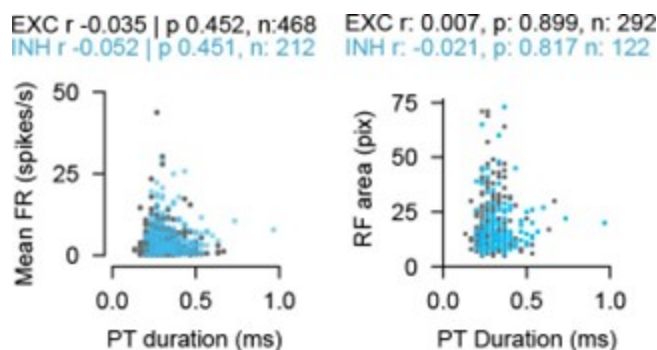
The GABAergic and nonGABAergic neurons showed a wide range of firing rates. It might be interesting to sort the cells by firing rates to see if they exhibit different properties. For example, since the SC contains both GABAergic interneurons and projection neurons it would be interesting to examine whether GABAergic neurons with higher firing rates exhibit narrower spikes, similar to cortical fast spiking interneurons. Similarly, it might be of interest to sort the neurons by their receptive field sizes since this is associated with different SC neuron types.

We thank the reviewer for the interesting suggestions of SC neurons classification into different categories. The relationship between connectivity measures and RF size has been addressed in Author response image 5. We have now studied the relationship of spike waveforms and several measures such as firing rate and RF size in more detail (Author response image 6).

As the baseline firing is generally low in SC and our experiments are performed under anesthetized conditions, we used the evoked firing rates to sort the cells by firing rates or RF sizes. We have added an analysis showing the mean firing rate (calculated over the full recording duration) as a function of the spike width (peak-to-trough duration). We observe no significant relationship between the different groups of cell types. The same accounts if we sort the SC neurons by their RF size. RF sizes were calculated from PSTHs and summed RF for SL and SD. We do not see a relationship between neuron type and firing or RF size.

Author response image 6.

Mean firing rate (left) and RF size (right) as a function of peak-to-trough (PT) duration for excitatory and inhibitory SC neurons. Both measures are not correlated to the PT duration (Pearson correlation coefficient, two-sided Wilcoxon rank-sum test).



The recording techniques allowed for the identification of the distance between connected retinocollicular fibers and postsynaptic neurons. It might also be interesting to compare the properties of connected pairs recorded at dorsal versus ventral locations since neurons with different genetic identities and response properties are located in different dorsal/ventral locations (e.g. Liu et al. Neuron 2023). Also, regarding the strength of connections, previous electron microscopy studies have shown that the retinocollicular terminals differ in density and size in the dorsal/ventral dimension (e.g. Carter et al JCN 1991).

We thank the reviewer for raising this interesting and relevant point to compare the properties of the connected pairs across the dorsal and ventral location. Unfortunately, our tangential recording approach is not ideally suited for comparing the properties of neurons across the different SC depths. For comparing dorsal versus ventral located neurons in the SC, as done in Liu et al., Neuron 2023, vertical recordings would be more appropriate. We now provide a discussion on this aspect (line 589).

Was optogenetic activation of GABAergic neurons ever paired with visual activation? It would be interesting to examine the receptive fields of the nonGABAergic neurons before and after activation of the GABAergic neurons (as in Gale and Murphy J Neurosci 2016).

This is an important point and indeed we have paired activation of GABAergic neurons with visual stimulation (checkerboard stimulus) to assess the impact of the GABAergic neurons on the firing of the excitatory neurons. We observed a diversity of effects, with some EXNs being strongly suppressed and others being only weakly suppressed. Thus, we predict that the receptive field of those EXN that are suppressed by optogenetically evoked IN firing, should be affected in some way. However, the checkerboard stimulus was only presented for a short duration (1 s) and for only a few trials (n = 30). Therefore, estimating the receptive fields of EXN before and after optogenetic activation of GABAergic neurons is unfortunately not possible with the existing dataset. We now mention this point in the discussion (line 668).

Reviewer #3 (Public Review):

This study performs in vivo recordings of neurons in the mouse superior colliculus and their afferents from the retina, retinal ganglion cells (RGCs). Building on a preparation they previously published, this study adds the use of optogenetic identification of inhibitory neurons (aka optotagging) to compare RGC connectivity to excitatory and inhibitory neurons in SC. Using this approach, the authors characterize connection probability, strength, and response correlation between RGCs and their target neurons in SC, finding several differences from what is observed in the retina-thalamus-visual cortex pathway. As such, this may be a useful dataset for efforts to understand retinocollicular connectivity and computations.

Recommendations:

Reviewer #1 (Recommendations For The Authors):

Some minor points.

Fig.1G shows a difference in mean firing rates between inhibitory and excitatory cells. Please plot the cumulative distribution of firing rates to be able to scrutinize the data better.

We have addressed this issue and updated panel G in Figure 1.

Fig. 2C. The black background color of this plot is black; it is not possible to decipher much, please change it to white

We have now changed panel C in Figure 2 to a white background.

Fig. 4D would be better represented as a histogram since most points overlap.

We now represent panel D in Figure 4 as a histogram.

Citations. I would cite some of the foundational work, in some instances, e.g., in the first sentence (SC receives input from the retina)

We have now addressed this issue and cited more foundational studies (e.g. line 68)

The discussion is a bit long; the last paragraph can be removed, mainly because the previous section conflates superficial SC with the entire SC, which is confusing (e.g., Ayupe et al.). In this way, there is more space to discuss the direct implication of the study within the context of known cell types.

We now shortened the discussion and provide more background about different SC cell types in the discussion (line 643).

Reviewer #2 (Recommendations For The Authors):

Minor correction: Whyland et al 2020 did not identify V1 input to horizontal cells. A more appropriate reference is Zingg et al Neuron 2017.

We thank the reviewer for this important point and have now corrected the citation in line 613 in the discussion to Zingg et al 2017.

Reviewer #3 (Recommendations For The Authors):

Regarding the degree of convergence from RGC to SC, the Crair lab (Furman 2013) performed a quantal analysis in slice that is worth citing.

We included this citation in the revised version of the manuscript (line 501).

I have lost track at this point, but many labs (Heimel, Meister, Farrow, Cang, Isa, maybe others?) have observed that neighboring SC neurons have similar tuning for direction/orientation, but the circuit mechanisms are not well understood. Given the relatively weak correlation between response tuning of RGC axons and their SC target neurons, a useful comparison might be that of SC neurons and their neighbors, and whether SC neurons that show weaker correlation to their RGC axons show stronger correlations with their SC neighbors, which could implicate local connectivity within SC.

We thank the reviewer for providing this interesting comment. With our recording approach we could study locally connected SC neurons. However, the focus of our study was to first characterize the retinocollicular connectivity and therefore investigating the intracollicular connectivity is beyond the scope of the current study. We thank the reviewer for the valuable suggestion and will consider to tackle this aspect in a separate study in the future.

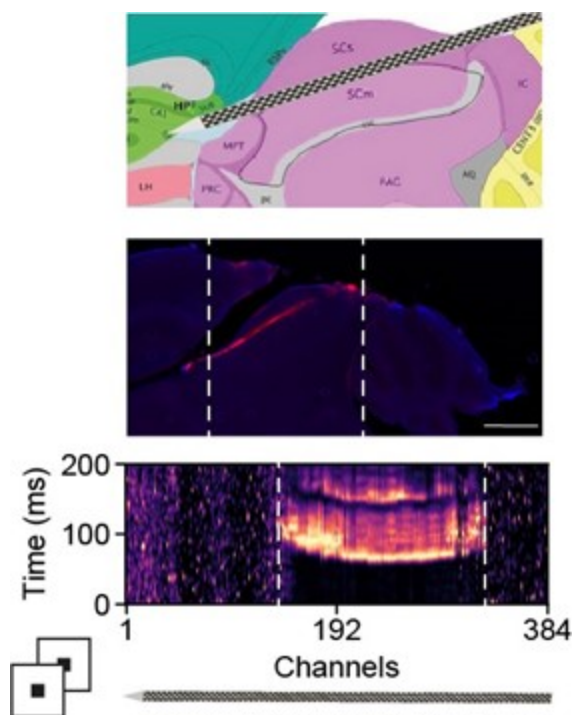
Is it possible any of these measurements are biased by laminar targeting of their probe within superficial SC? Their schematic seems to suggest they targeted the deeper part of

superficial SC. Do they know whether they recorded throughout superficial SC or targeted the deeper layers closer to stratum opticum?

Our recordings are in between the deeper and upper visual SC layer depending on the recording site on the Neuropixels probe as we use an angled insertion approach. Besides DiI staining (Author response image 7), we can estimate the location of the probe using functional measurements, i.e. visually driven channels and retinotopic locations of the recording sites. If the Neuropixels probe is inserted too superficial, the number of recording site with visually driven activity is low. If the Neuropixels probe is inserted too deep in the visual layers we see two separated regions on the probe with visually driven activity in which the retinotopy is non-continuous (please refer to Figure 2 in (Sibille et al., 2022)). In the recordings included in this study, the number of visually driven channels was generally high and the retinotopy continues, suggesting that we covered a region within the deeper and upper visual layers.

Author response image 7.

Functional estimation of probe location. DiI staining of Neuropixels probe (middle) and multi-unit activity across channels in response to visual stimulation (bottom). The white dashed lines in the middle and bottom panels mark the rough boundaries of the visual SC layers.



In Fig. 4, the authors argue that firing in inhibitory neurons is less correlated with RGC input. Does their metric for contribution of retinal input control for the fact that inhibitory neurons have higher firing rates overall and, e.g., may be more depolarized at rest and likelier to fire spontaneous spikes but no less likely to be driven by retina? Or is the argument that their visual responses are more likely to be driven by V1 or local connections?

We thank the reviewer for bringing up that point. The contribution measure estimates the fraction of SC spikes that were preceded by an RGC spike and it is thus, in theory, independent

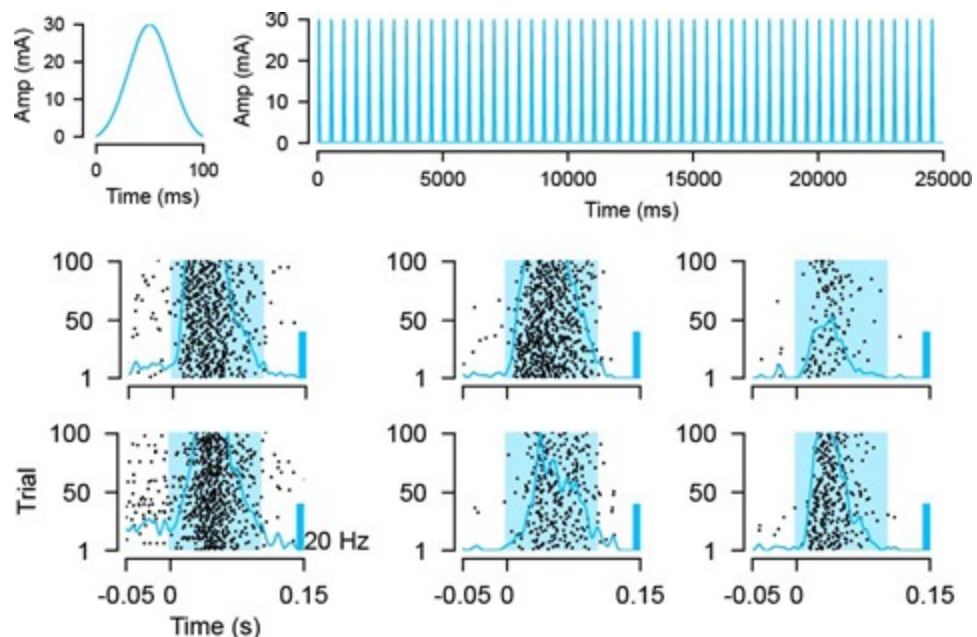
of the firing rate of the SC neuron. In practice, however, we agree that high firing SC neurons may be more likely to have a lower contribution value simply because a larger fraction of their spikes is not preceded by the activity of the presynaptic RGC. But this is exactly what we aimed at characterizing with this analysis. Where these non-RGC driven SC spikes originate from, whether from a more depolarized state of the neuron or by other sources such as V1 or local connections, we can only speculate about. That said, please note that despite SC INs having higher firing rates, not all of them show low contribution. Likewise, we also see SC neurons with low firing rates and low contribution values (new Supp Fig. 3).

Minor point: The optotagging in the example cell doesn't cause the cell to fire for ~50 ms? That is odd. Typically, cells classified as optotagged fire within 5-10 ms of light onset. Is that a strange example cell or is there something different about the optotagging approach?

Unfortunately, transient LED light onsets and offsets can induce light artifacts on Neuropixels probes (Jun et al., 2017; Steinmetz et al., 2021) and therefore it is challenging to use brief LED pulses for optotagging with Neuropixels probes. To avoid this overlap of artefacts and LED evoked spikes, we opted for a longer stimulus duration of 100 ms to activate VGAT neurons (Bennett et al., 2019; Siegle et al., 2019). Moreover, instead of a square pulse, we used a slow ramping for light onsets and offsets to minimize the magnitude of induced artifacts. In Author response image 8 we present examples of individual activated VGAT neurons responding to a 100 ms blue light pulse.

Author response image 8.

Optotagging approach. Example traces of a single stimulation pulse and protocol used for optogenetic stimulation. Evoked activity in response to LED stimulation (100ms, 100 trials) for six example SC IN neurons.



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